Why Randomization Works

Initialise

Initialise

Potential Outcomes

Initialise

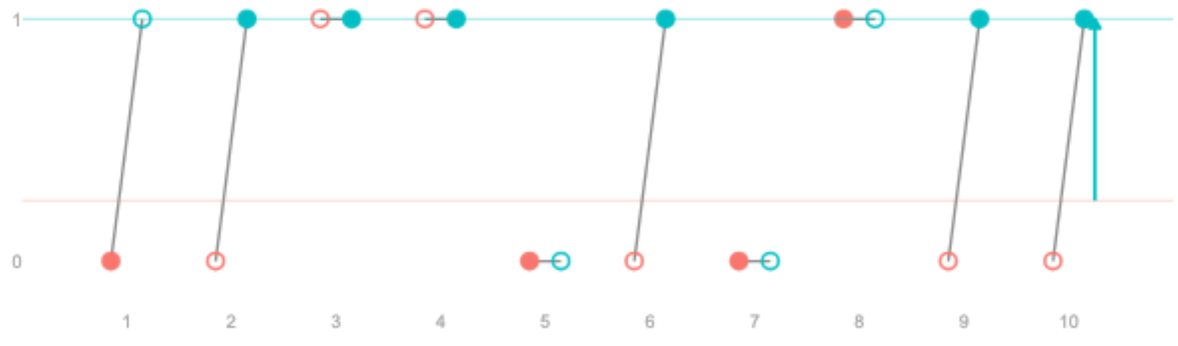
Survey vs. Experiment

The survey



You chose the dark chocolate. We asked three from each group.

The experiment



A shuffle chose who got the coconut milk chocolate. We asked everybody.

- Both gave us two groups of ratings and the same estimator, the difference in group means.

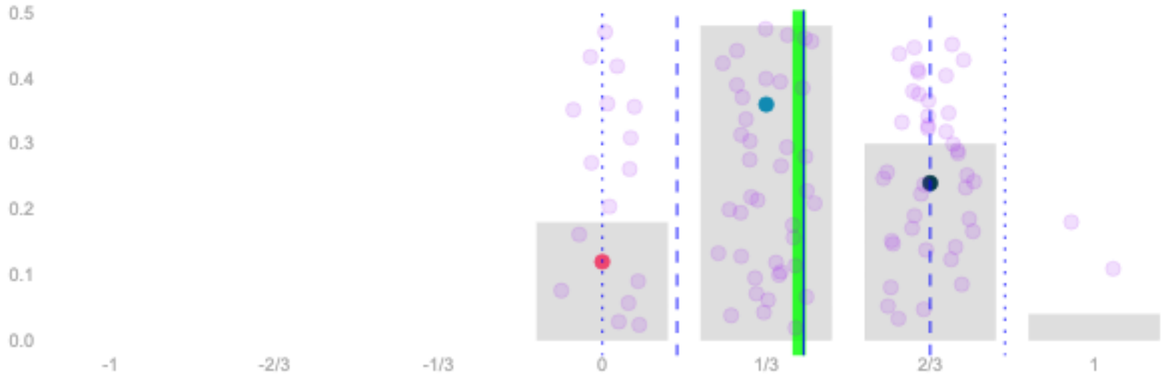
$$\hat{\Delta} = \hat{\mu}(1) - \hat{\mu}(0)$$

- What varies is different in the two.
  - → In the survey, the sample: a different three from each group gives a different $\hat{\Delta}$.
  - → In the experiment, the assignment: a different shuffle gives a different $\hat{\Delta}$.
  - → Step forward and the two distributions of $\hat{\Delta}$ look much the same.
- A dot we never saw is drawn hollow.
  - → In the survey, that's the people we didn't ask.
  - → In the experiment, that's the rating you'd have given the other chocolate.
- So we never saw more than one rating per person, even when we asked everybody.
- *Today we'll ask ourselves:* when does $\hat{\Delta}$ tell us about an effect, and what does shuffling have to do with it?

Initialise

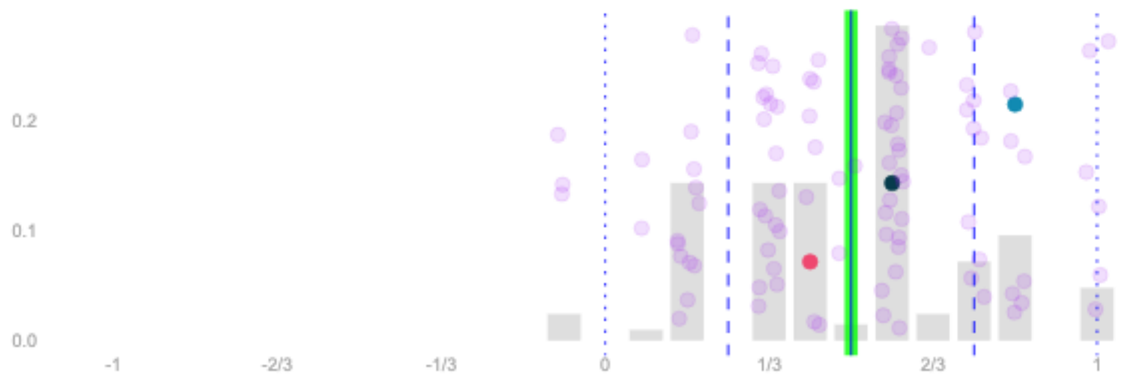
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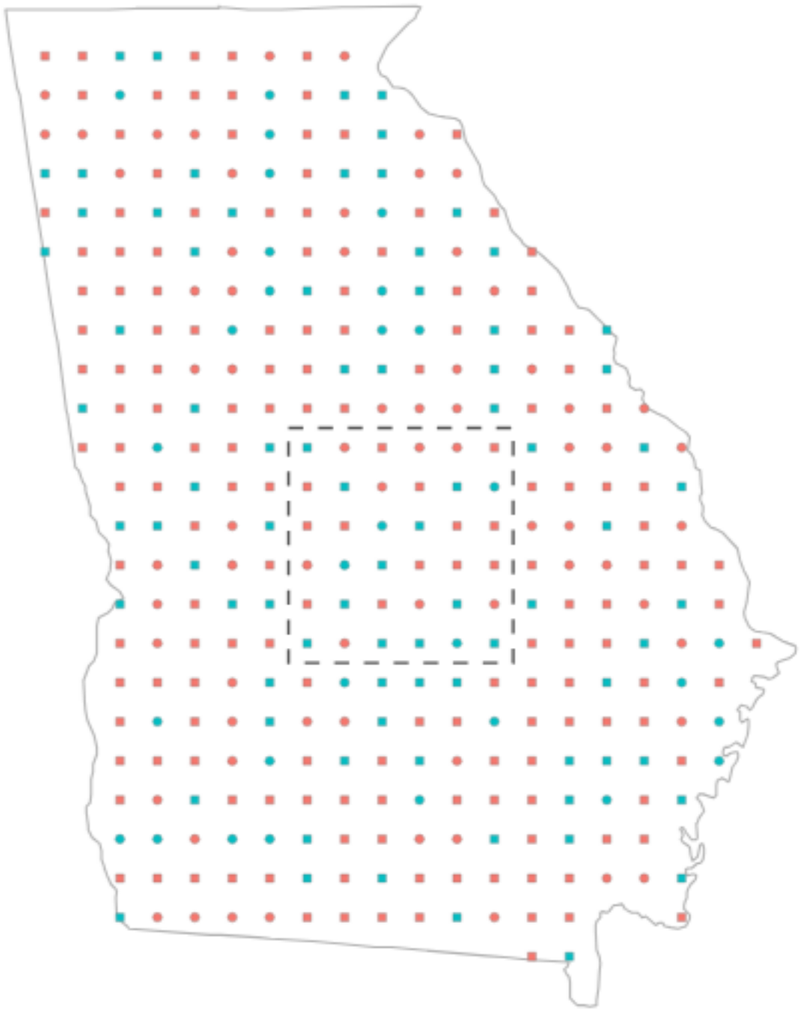
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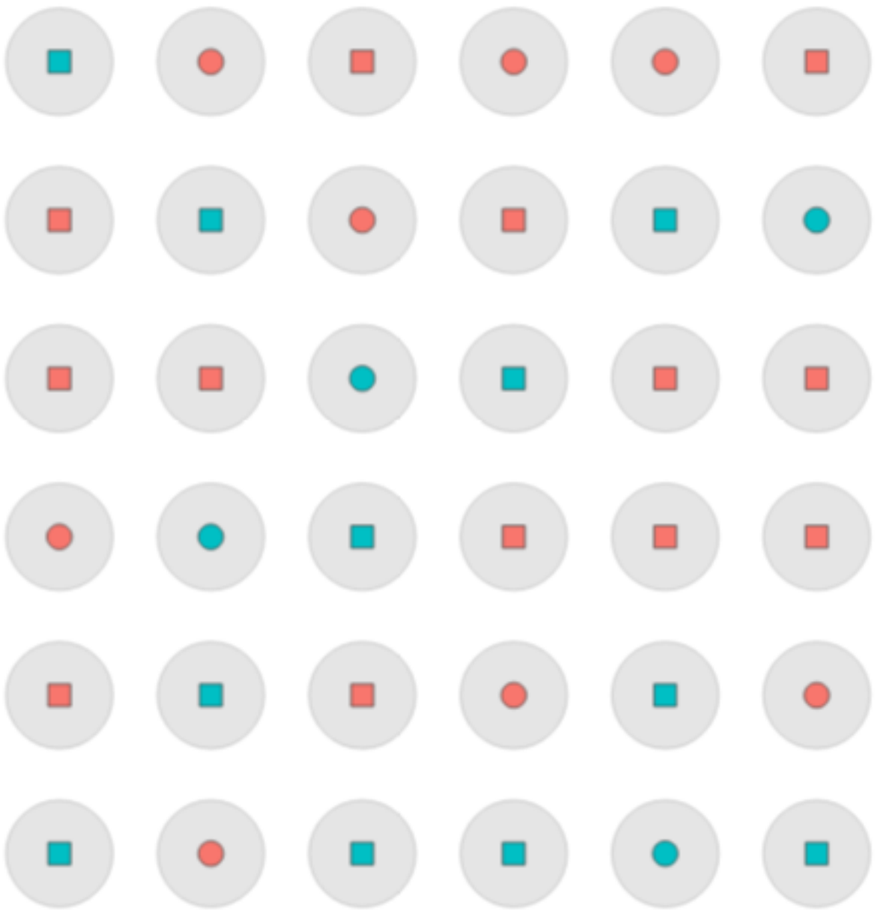
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Initialise

Black vs. non-Black Voters in GA

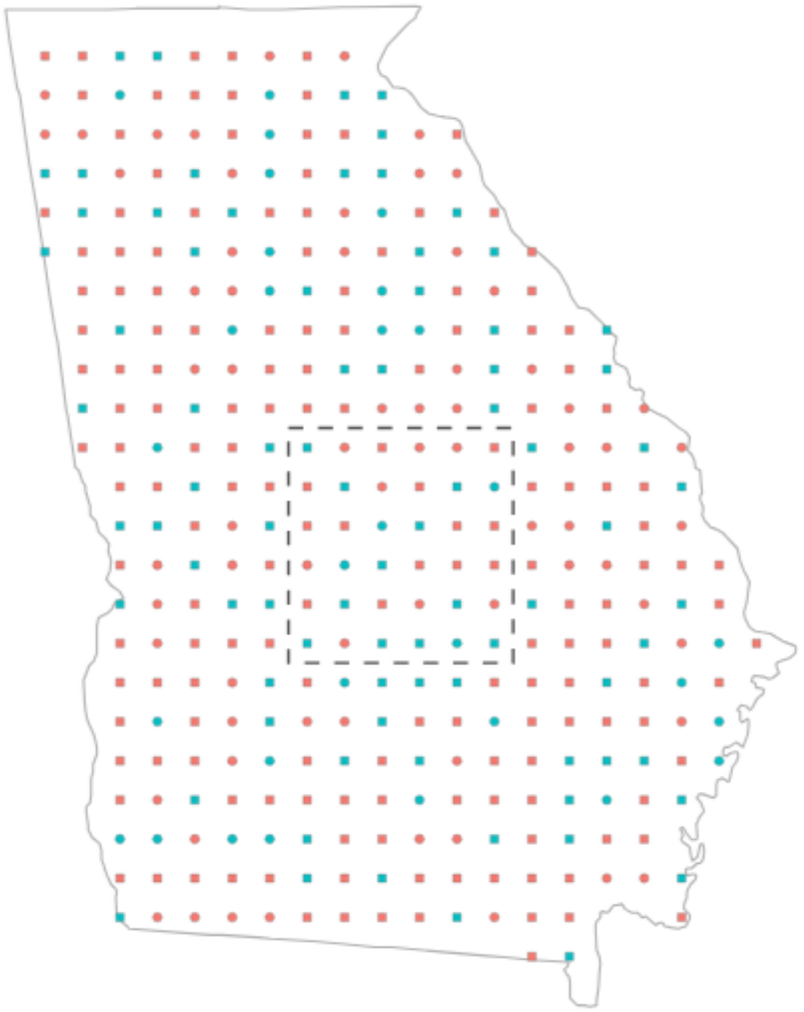


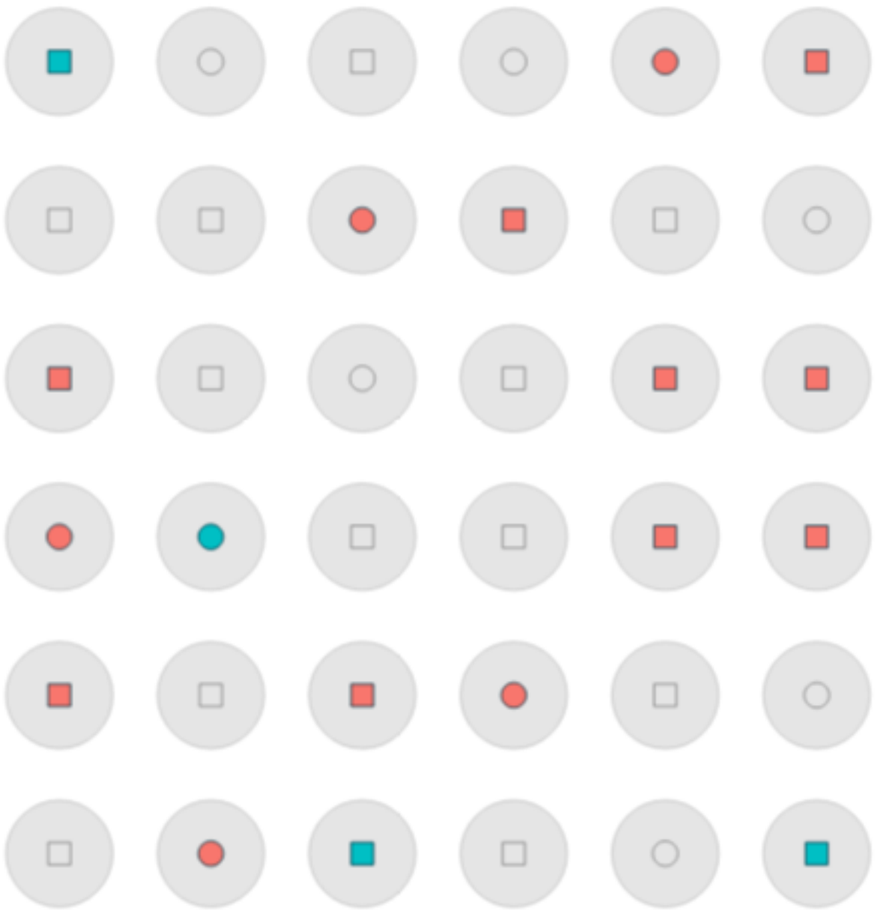


- Every voter in our population belongs to one group ($w_j = 1$ or $w_j = 0$) and has one turnout status.
  - → ■ voted
  - → ● didn't vote
- We draw a random sample of voters from our population.
  - → The voters we drew are filled in; the ones we didn't are hollow.
- Each draw gives a different sample — and a different $\hat{\Delta}$.

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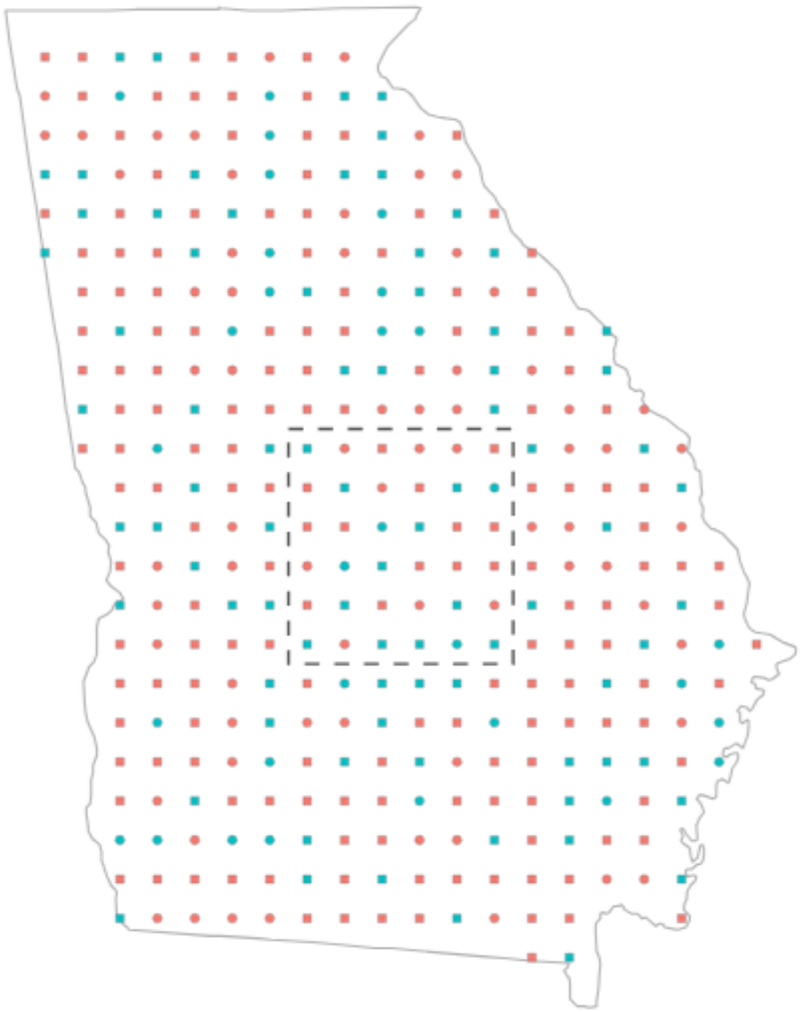


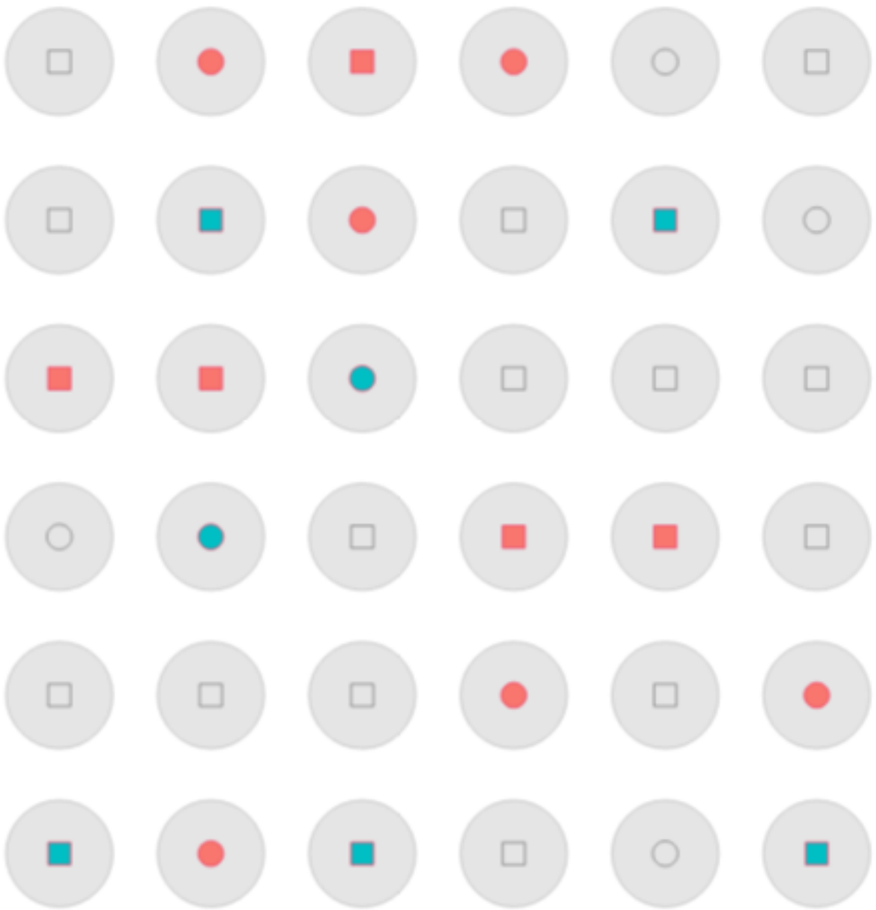


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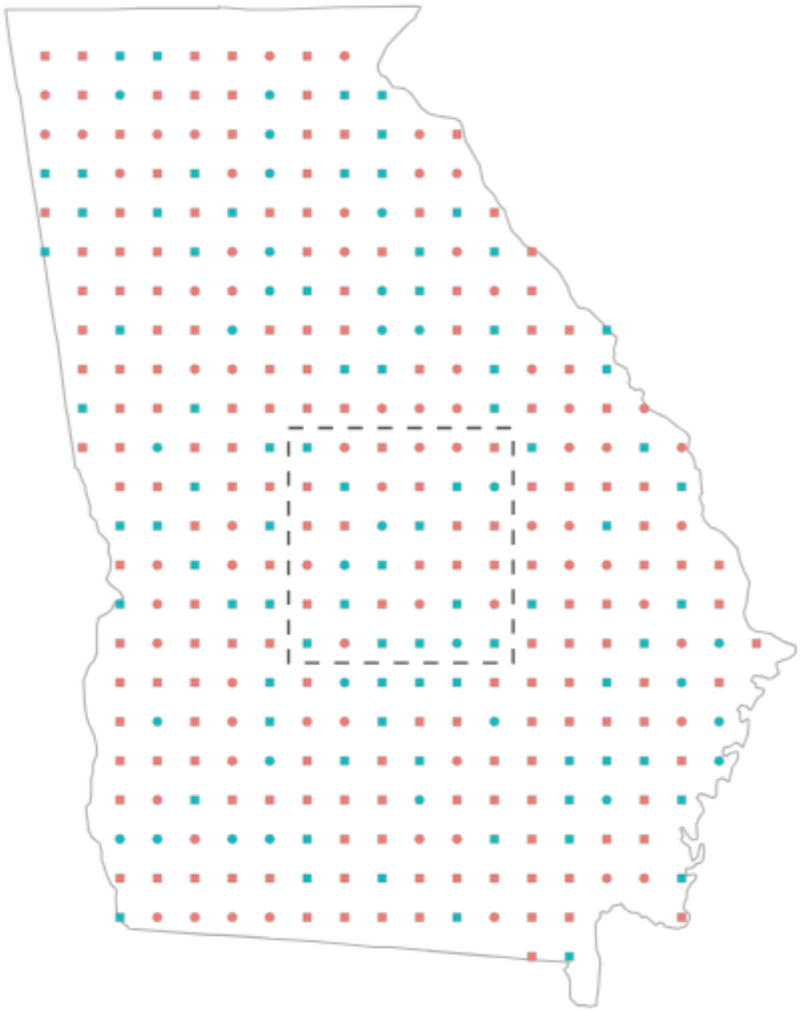


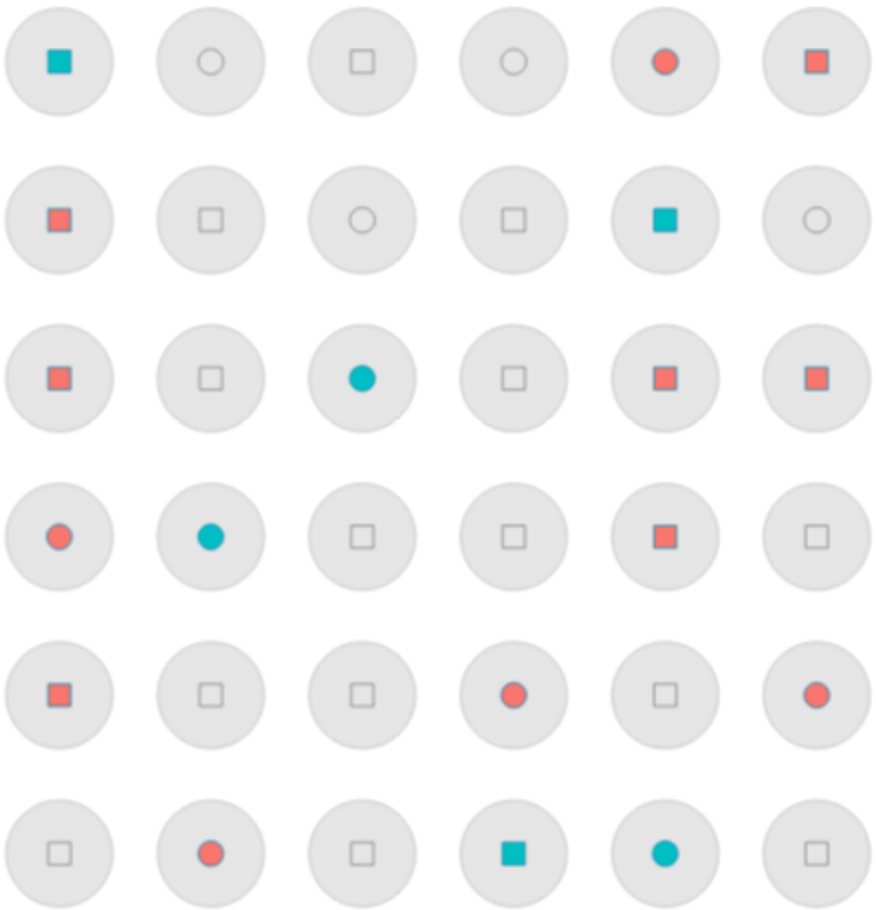


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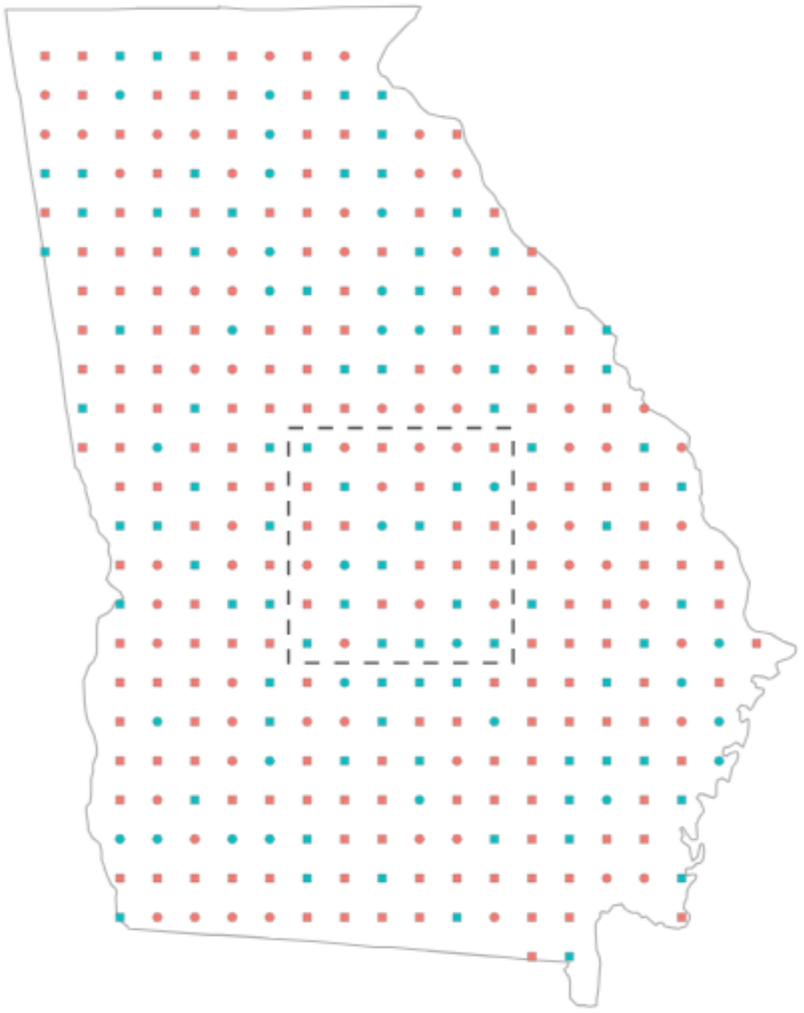


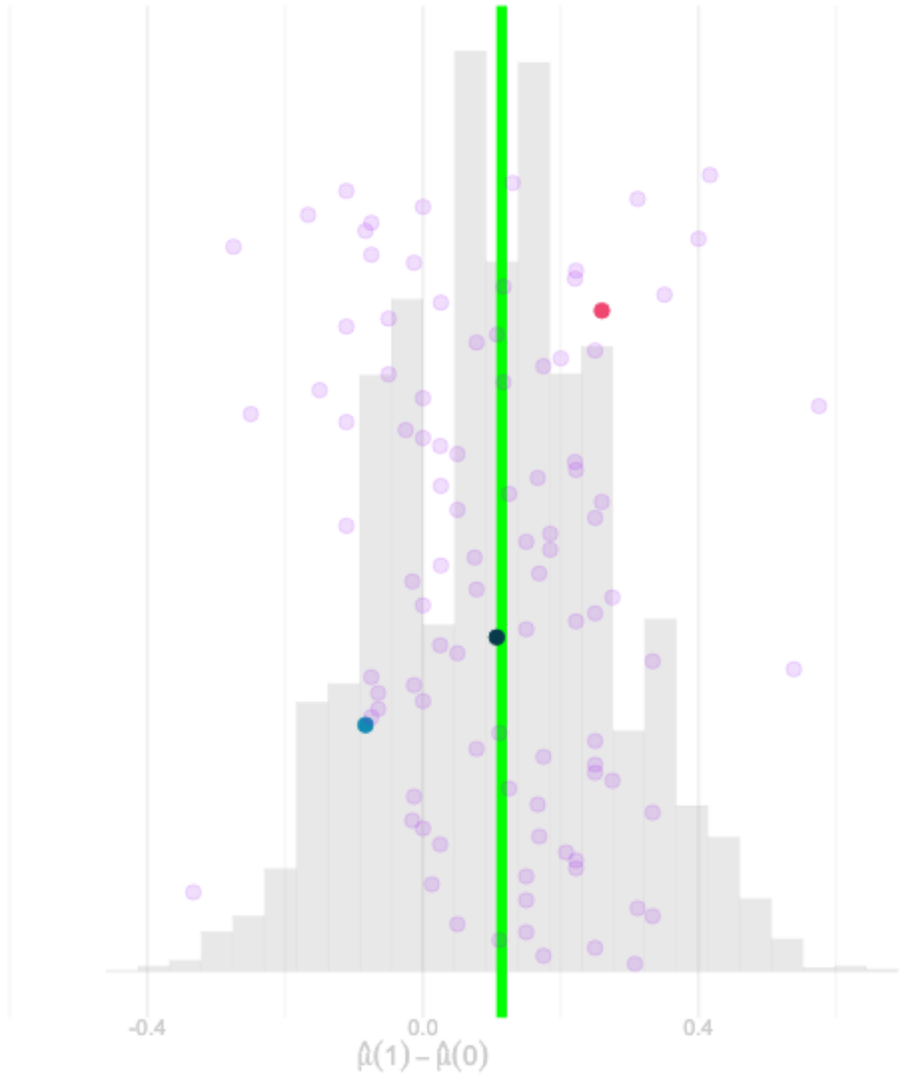


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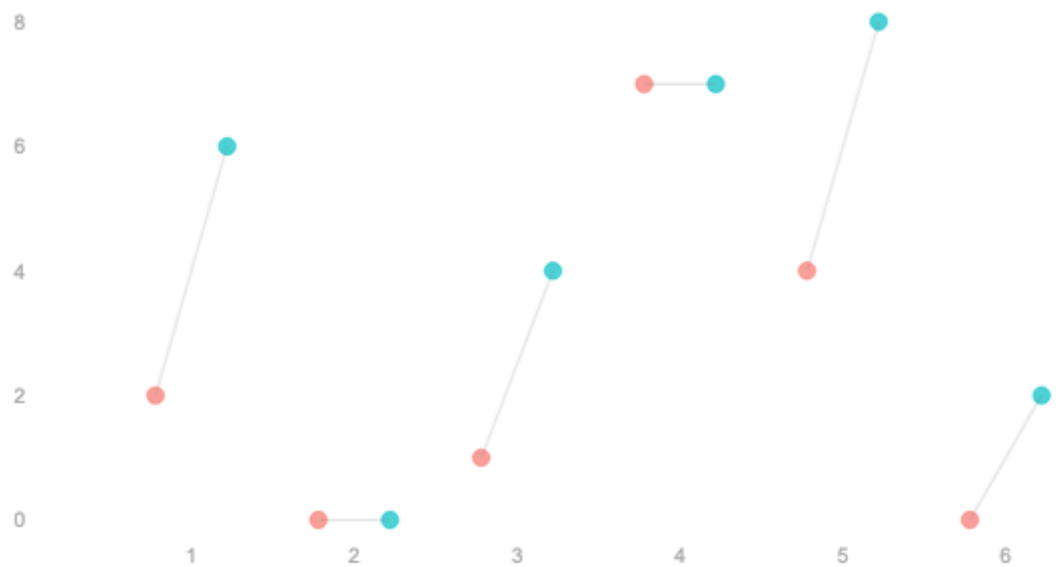




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Initialise

The Potential Outcomes Formalism



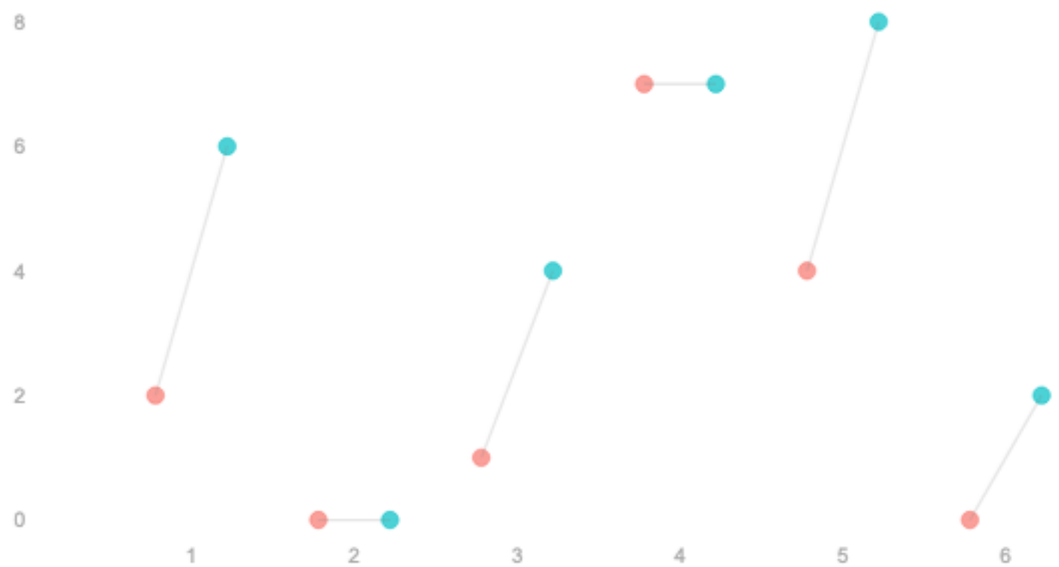
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1	6	2	4
2	0	0	0
3	4	1	3
4	7	7	0
5	8	4	4
6	2	0	2

Six participants in the National Supported Work (NSW) job-training experiment, with 1978 earnings in thousands of dollars.

- To reason about cause and effect formally, we use **potential outcomes**.
  - → $y_j(0)$ is the amount person j would earn if they were not in the job program.
  - → $y_j(1)$ is the amount person j would earn if they were in the job program.
- *Terminology*. We call the actions we can take **treatments**.
  - → Each individual in our population has a potential outcome for each treatment.
  - → In this case, we have two treatments and therefore two potential outcomes—a pair—for each individual.
  - → Each individual's potential outcomes are drawn as a connected pair of dots, one red and one green.
- How much higher would average earnings be if we treated everyone vs. if we left everyone in control?
- *Question*. How do we write our target in terms of these potential outcomes?

Initialise

The Potential Outcomes Formalism



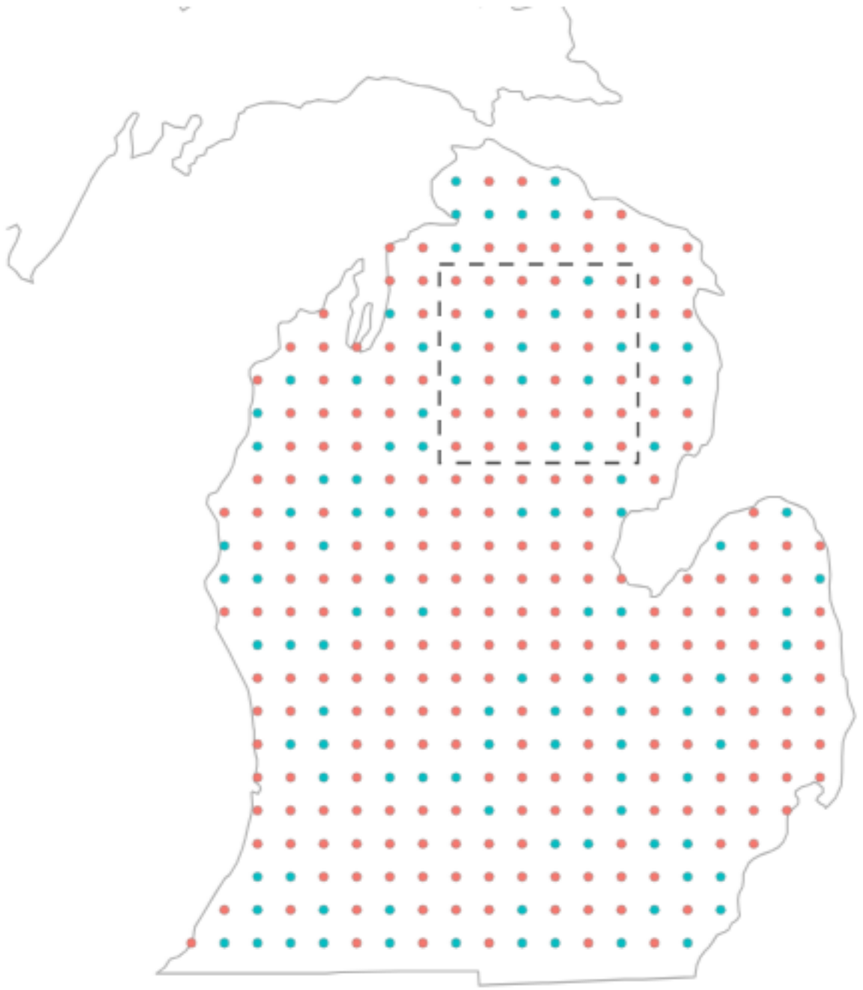
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Initialise

Recipients of Threatening Letters vs. Others in MI

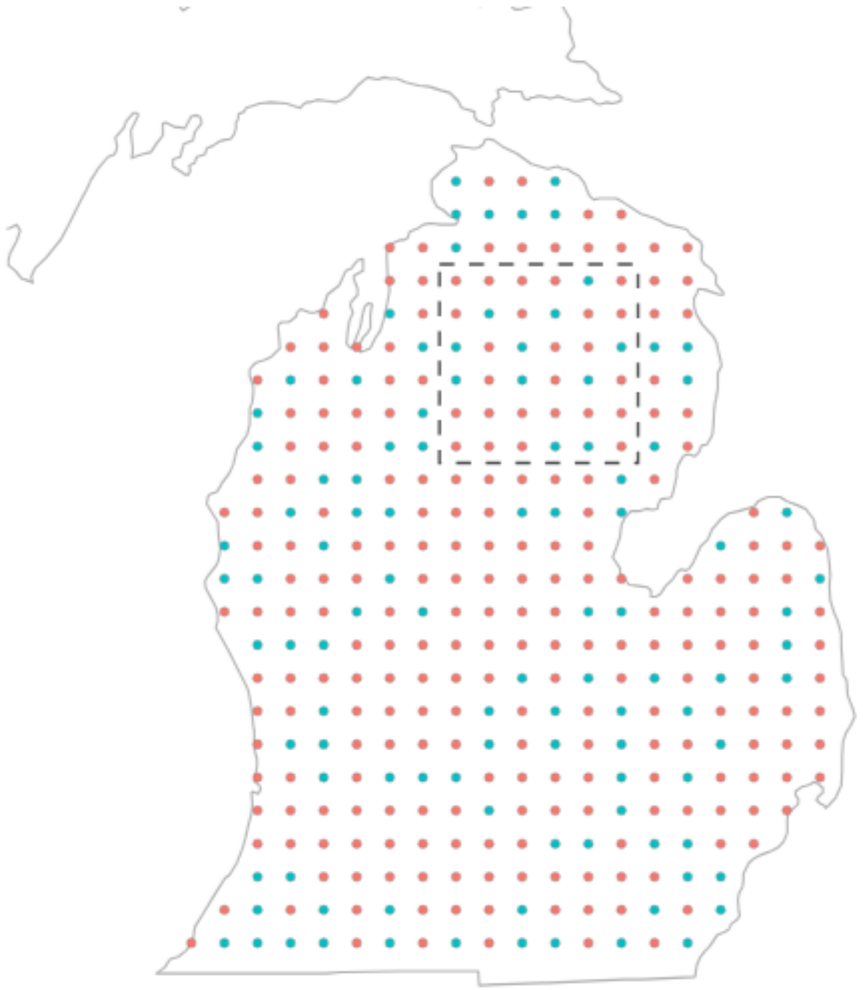


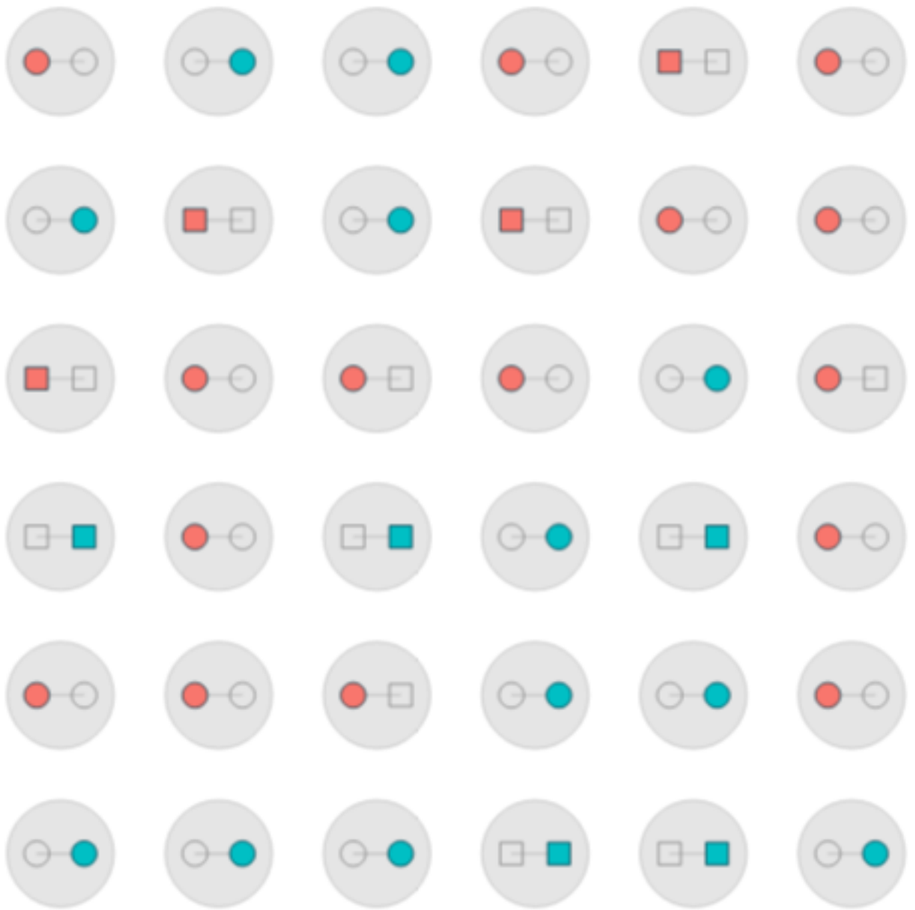


- Our population is the households in the study, not the whole state.
- We randomize which of them get no mailer or mailer.
  - → Each household has two potential outcomes, $y_j(0)$ on the left and $y_j(1)$ on the right.
    - ■ it votes under that treatment
    - ● it doesn't
- Randomizing fills in the outcome we see.
  - → The hollow one is the counterfactual.

Initialise

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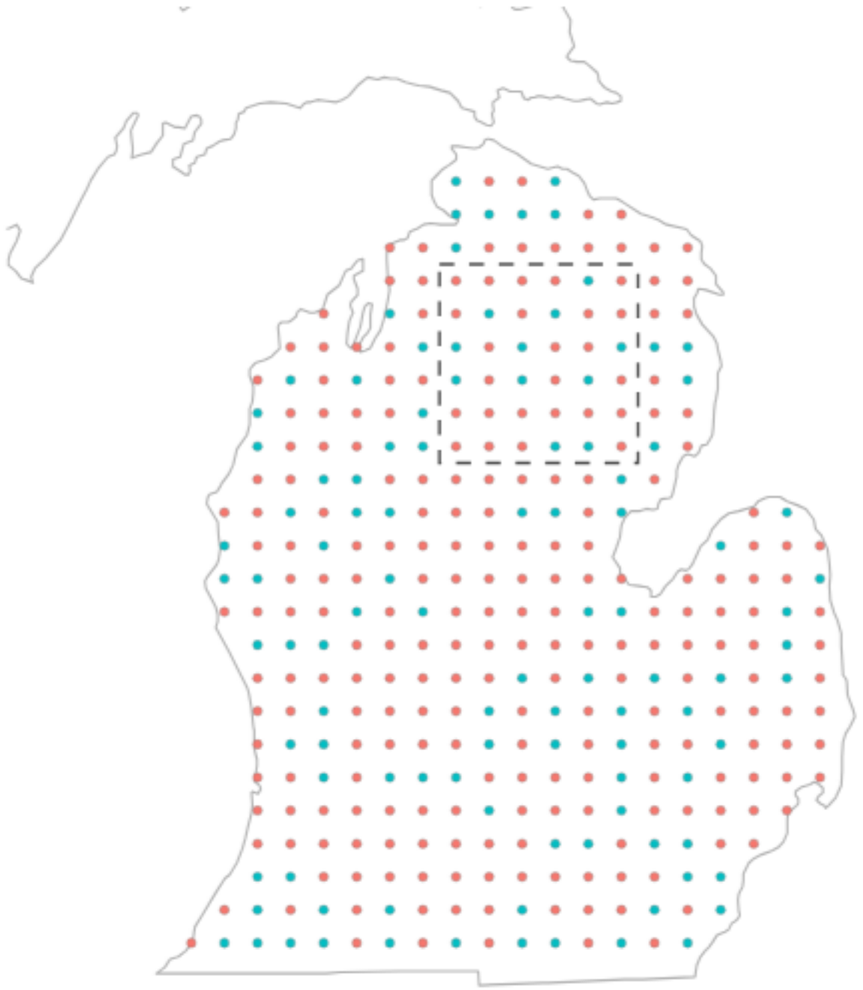




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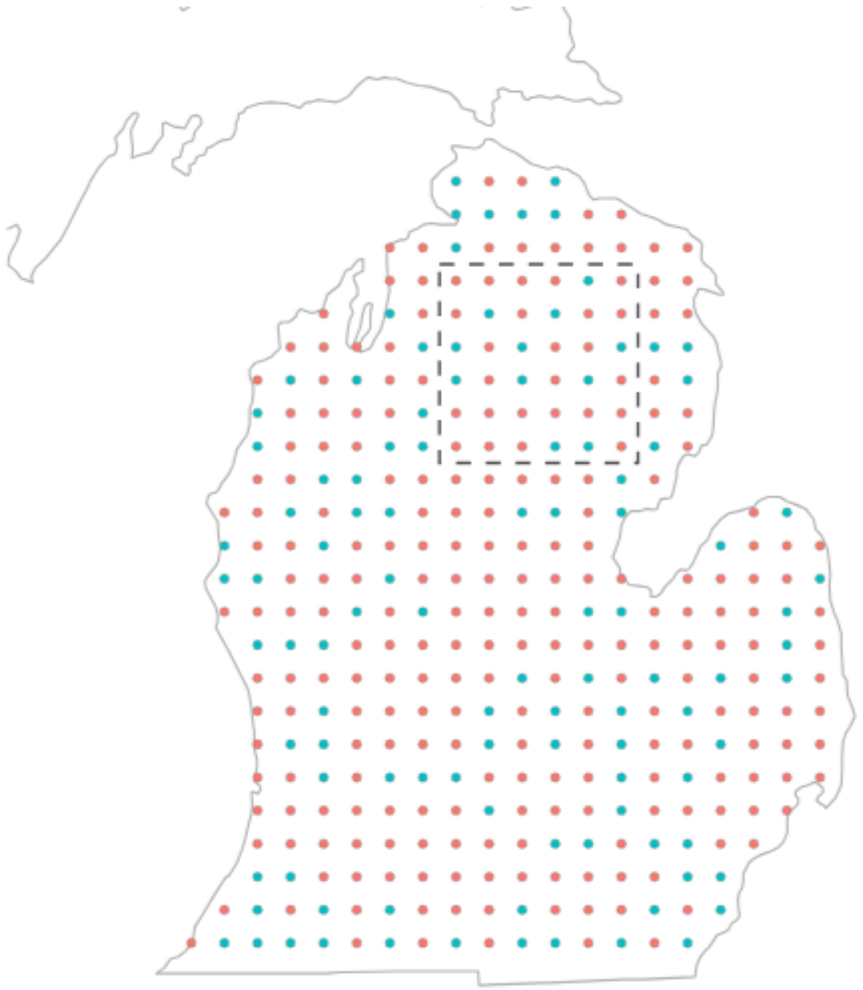


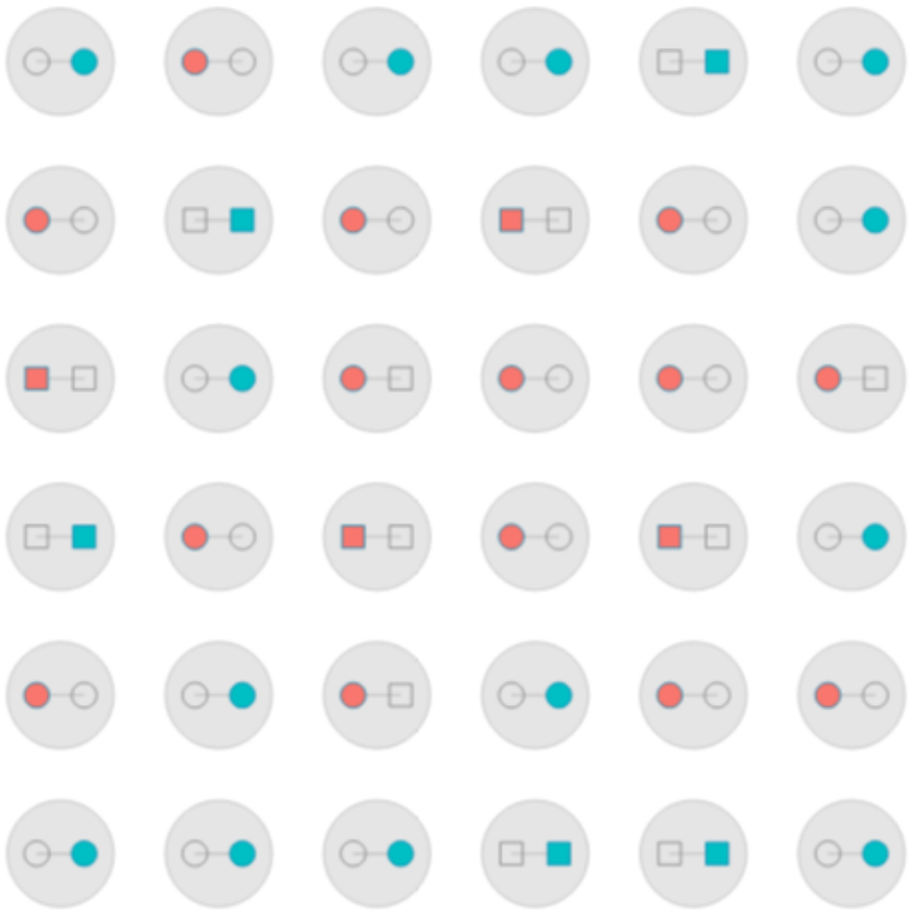


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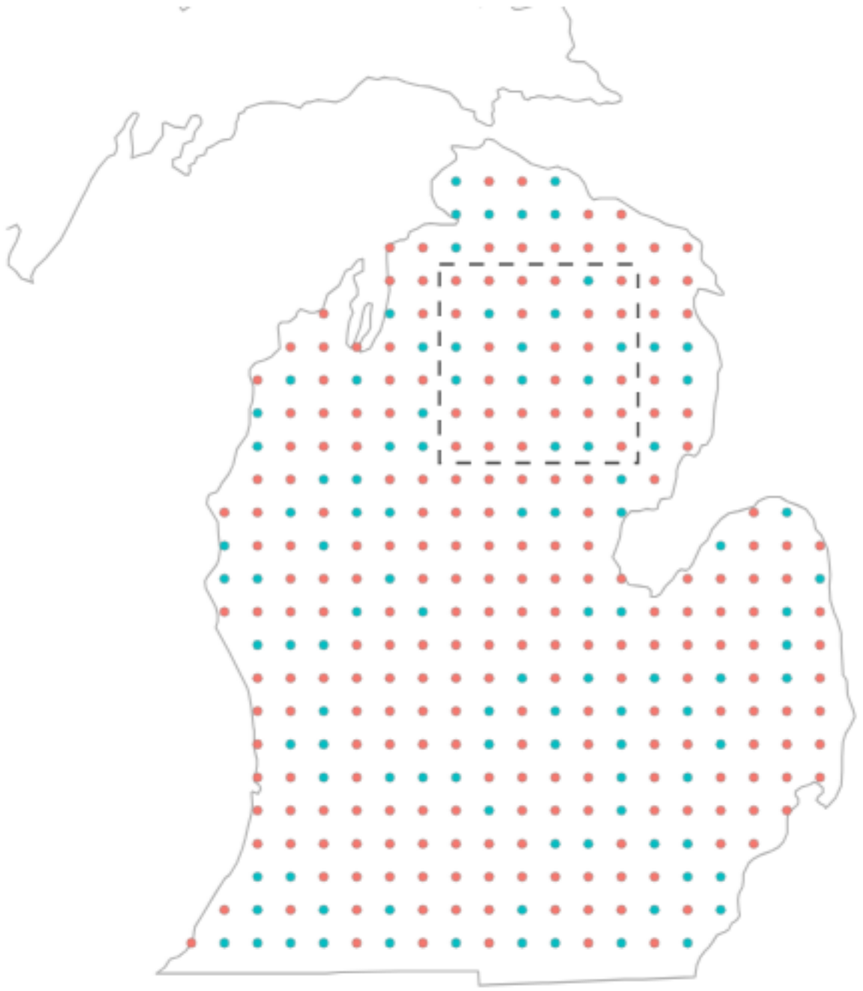


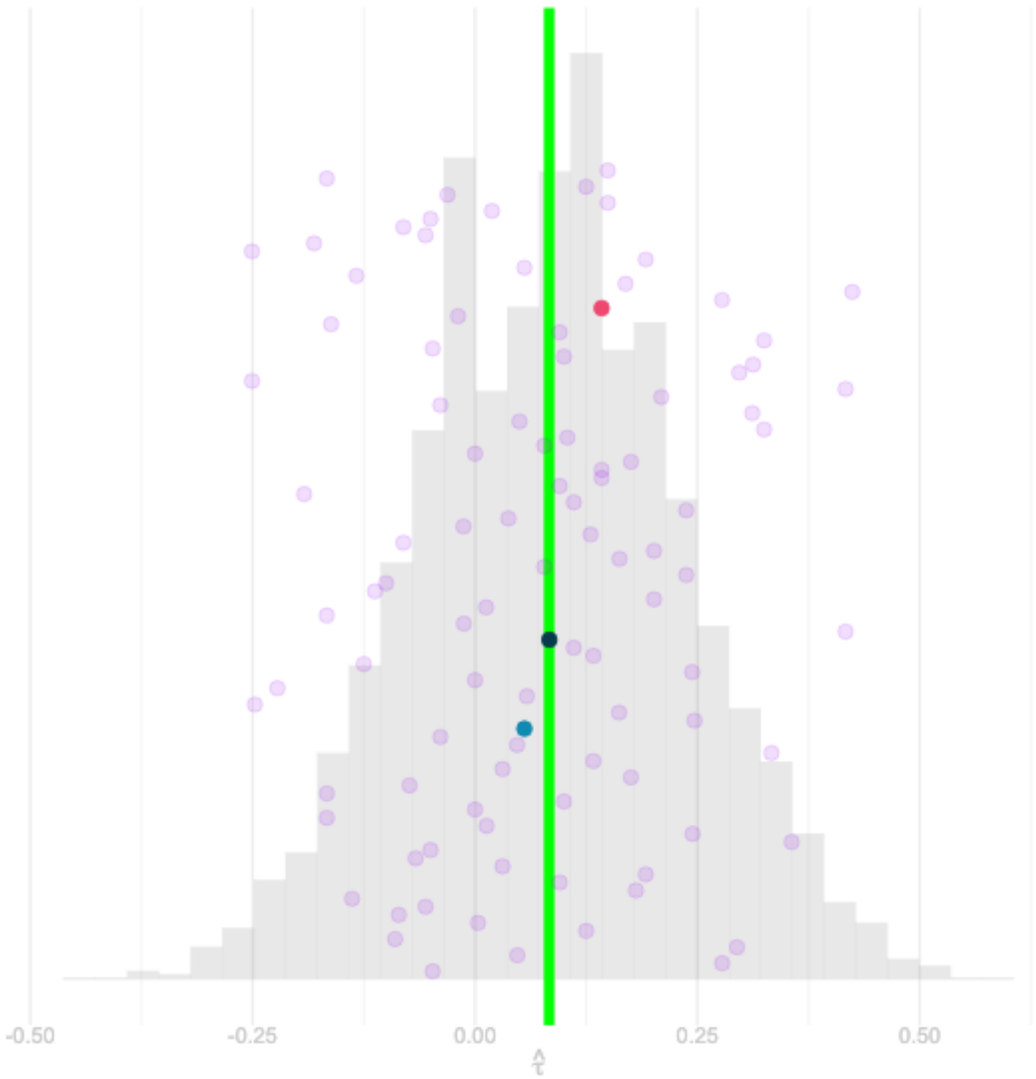


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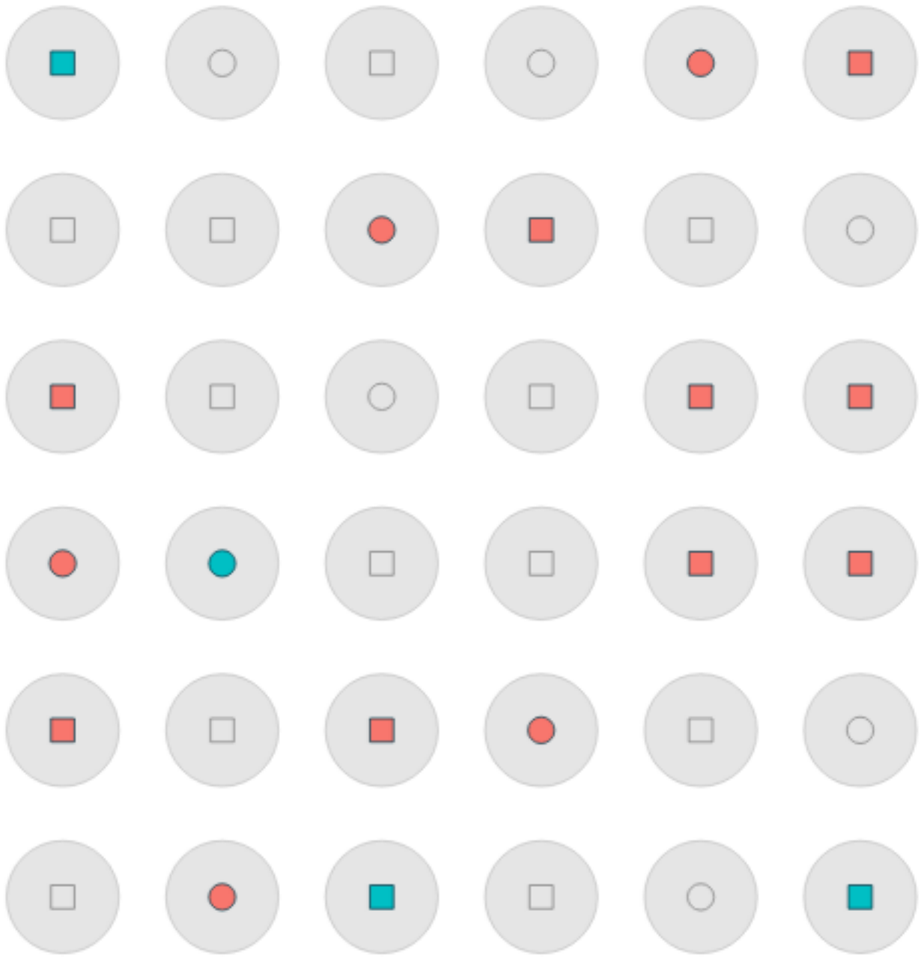




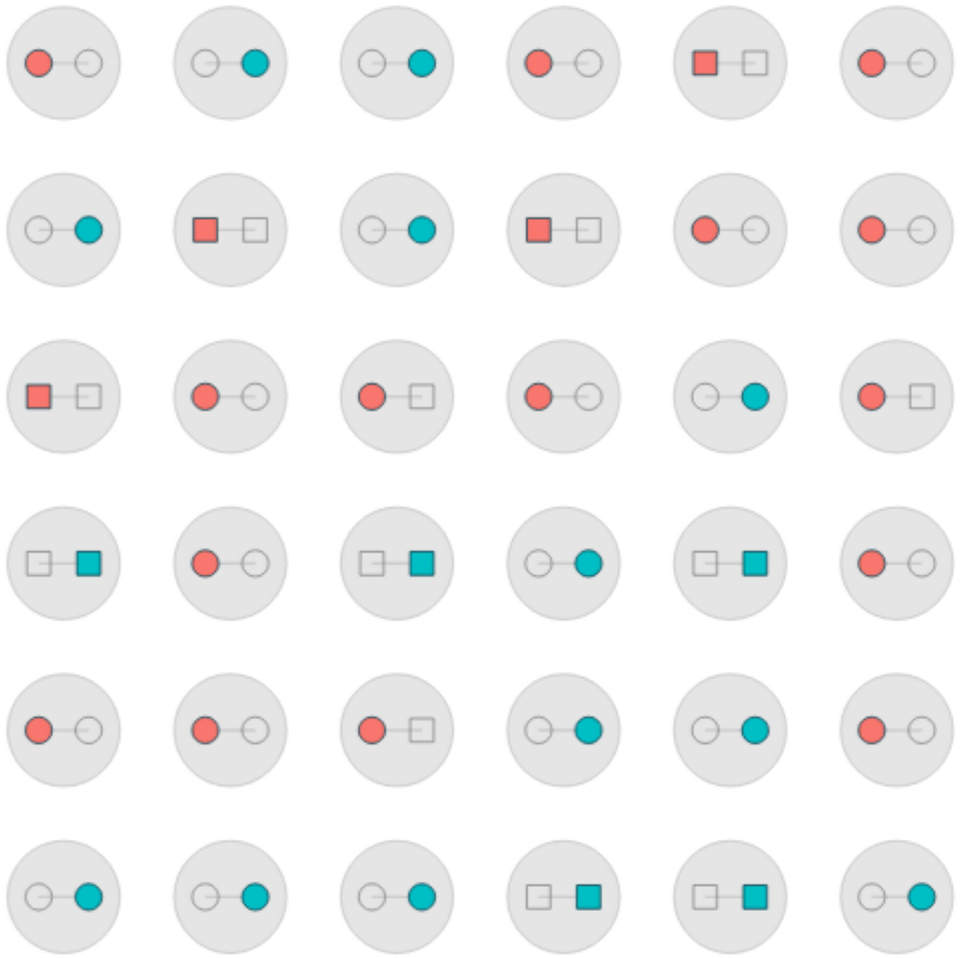
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Initialise

Sampling vs. Randomization



Georgia: sampling chooses which voters we see



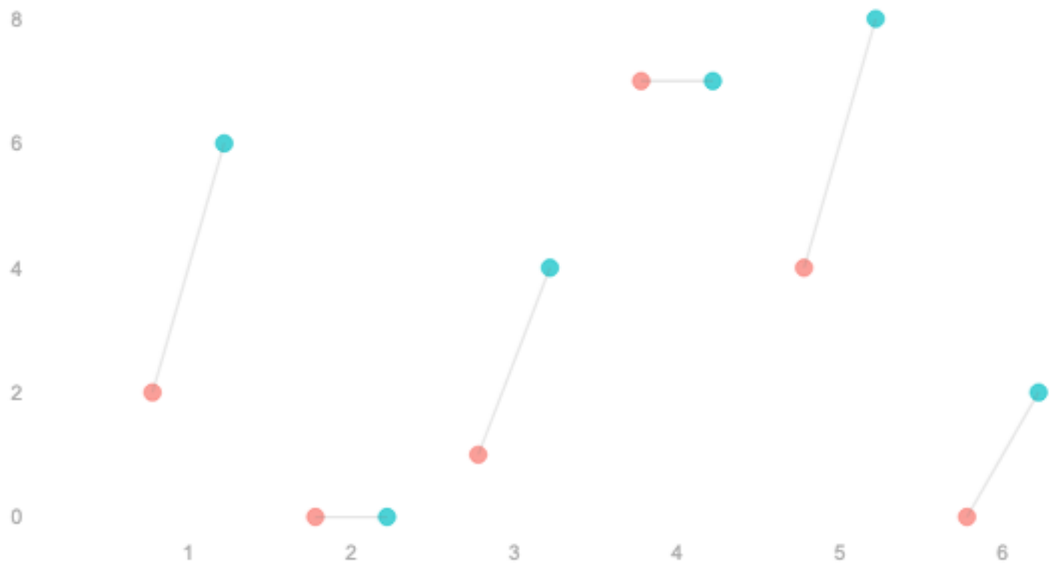
Michigan: randomization chooses which of two potential outcomes we see

- Both give us two groups of turnout numbers and the same estimator $\hat{\Delta}$.
- The mechanisms are not the same.
- Nor is what they estimate.
  - → Georgia compares two groups of people.

$$\Delta = \mu(1) - \mu(0)$$

Initialise

Potential Outcomes as Functions



j	$y_j(1)$	$y_j(0)$	τ_j
1	6	2	4
2	0	0	0
3	4	1	3
4	7	7	0
5	8	4	4
6	2	0	2

- It's convenient to think of each individual's potential outcomes as *one function* instead of *two values*.
  - → $y_1(\cdot)$ is the function defined like this.

$$y_1(w) = \begin{cases} 6 & \text{if } w = 1 \\ 2 & \text{if } w = 0 \end{cases}$$

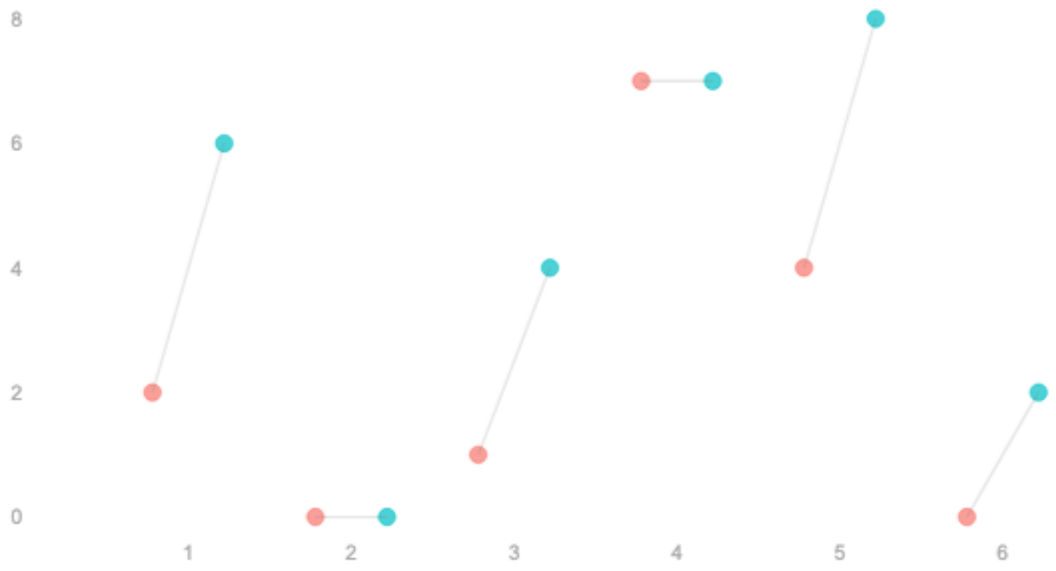
  - → $y_j(\cdot)$ is the function defined like this. The only way the notation really lets us define it.

$$y_j(w) = \begin{cases} y_j(1) & \text{if } w = 1 \\ y_j(0) & \text{if } w = 0 \end{cases}$$

  - → y_j is a function of w. That's it. $y_j(w)$ is a constant.
- That individual's **realized outcome** is …
  - → the value $y_j(w_j)$ their potential outcome function returns
  - → when we plug in the treatment w_j that they actually receive.

Initialise

Potential Outcomes as Functions

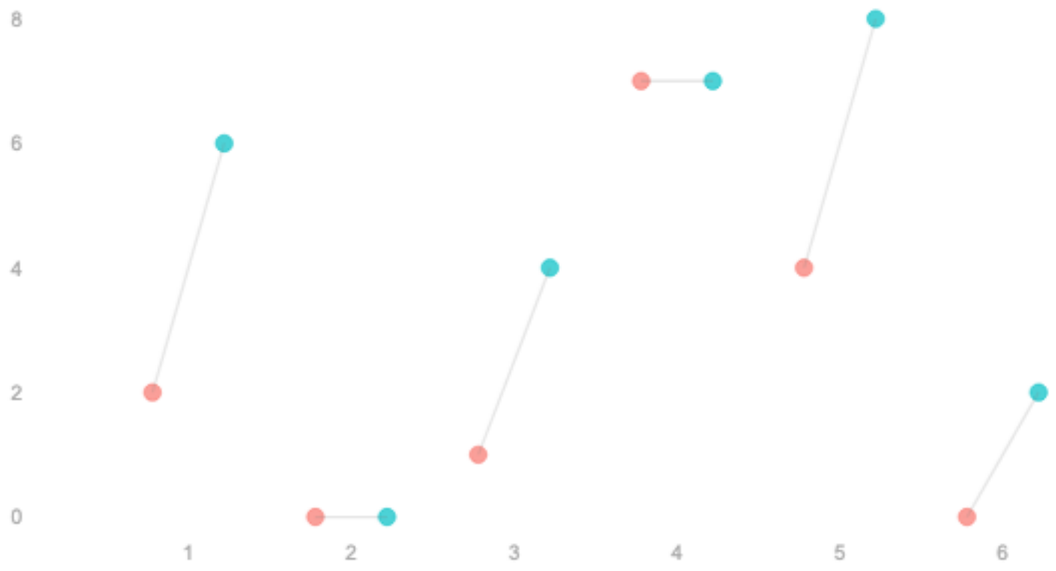


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- Suppose $\boldsymbol{W}$ is a coin flip: a random variable taking on the values $\mathbf{0}$ and $\mathbf{1}$ with equal probability.
- Write out a table describing the joint distribution of $\boldsymbol{W}$ and $\boldsymbol{y_3(W)}$.
- Step forward to see the solution.

Initialise

Potential Outcomes as Functions



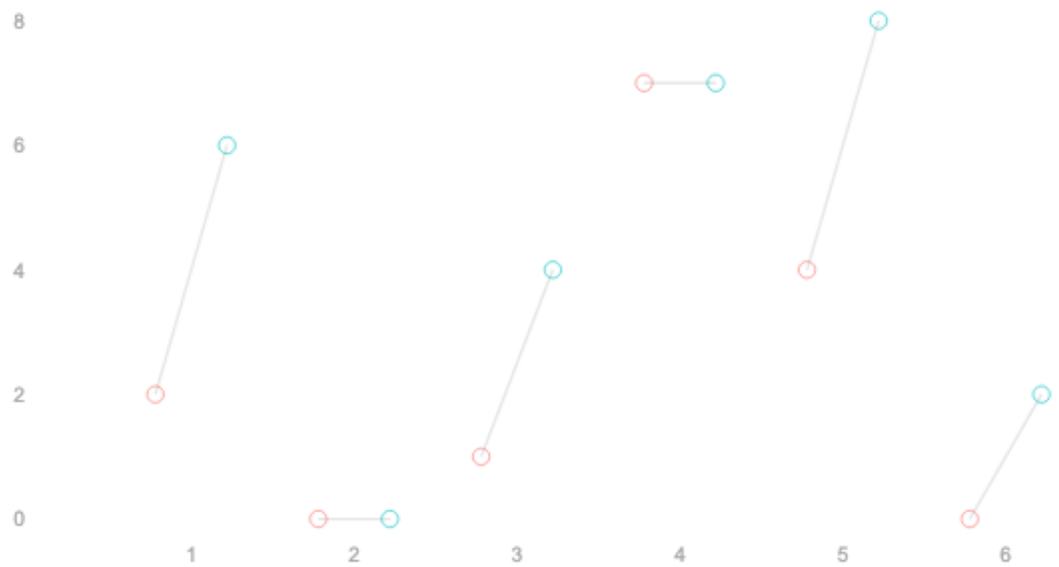
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$\mathbf{probability}$	W	$y_3(W)$
$1/2$	0	1
$1/2$	1	4

Initialise

The Fundamental Problem of Causal Inference

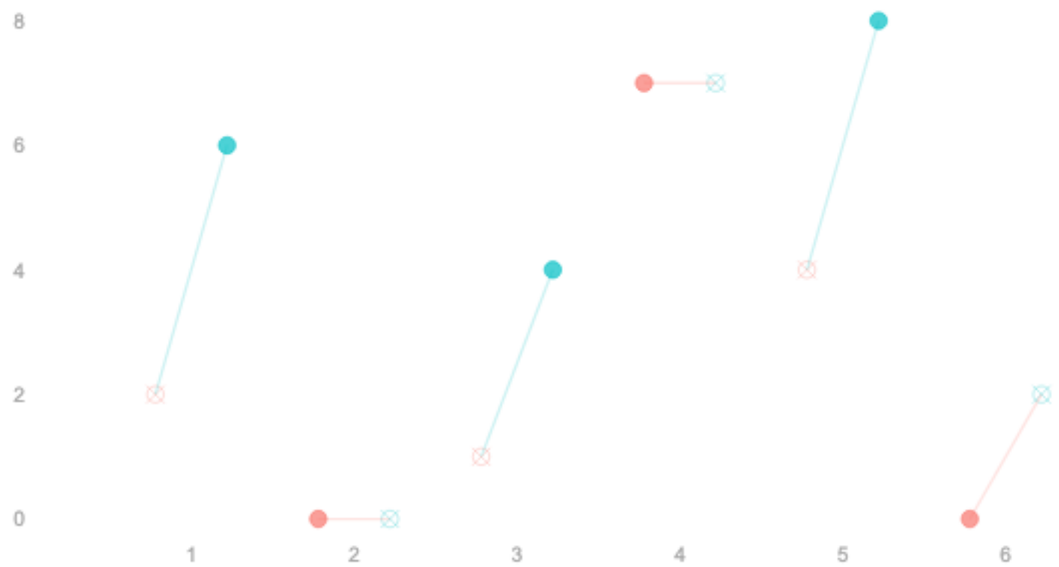


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- In concrete terms.
  - → You can't both put someone in the program and not put them in the program.
  - → So you can see what happens when you treat someone or what happens when you don't, but not both.
- In abstract terms.
  - → Each individual can take only one treatment.[1]
  - → So only one of each individual's potential outcomes is realized.
- That means that, even if we can choose everyone's treatment however we want:
  - → We can't calculate anybody's *individual treatment effect* $\tau_j = y_j(1) - y_j(0)$.
  - → And we can't calculate the average of them, $\bar{\tau} = \frac{1}{m}\sum_{j=1}^{m} \tau_j$, either.

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The Fundamental Problem of Causal Inference

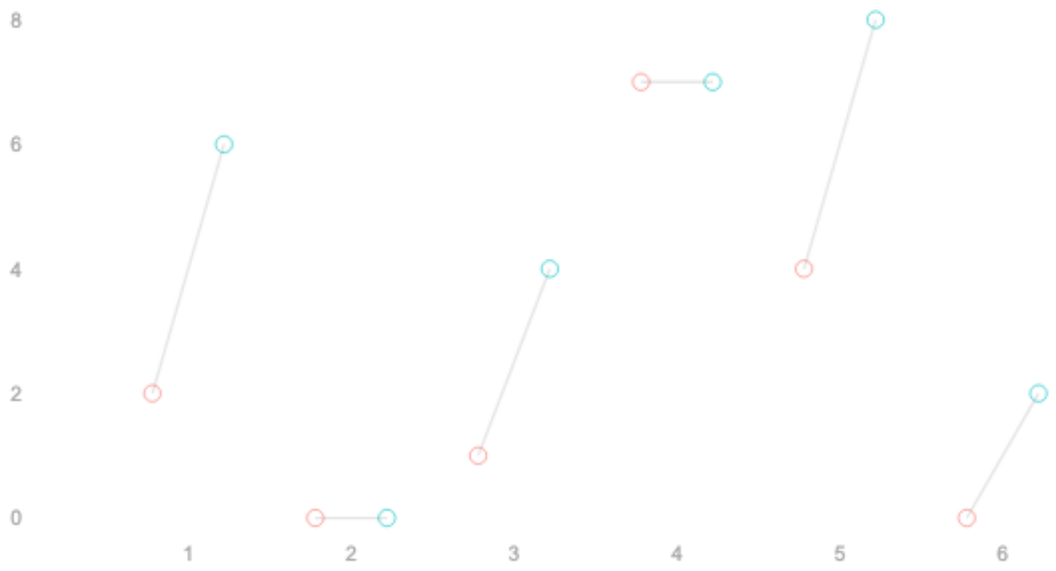


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Initialise

The Fundamental Problem of Causal Inference

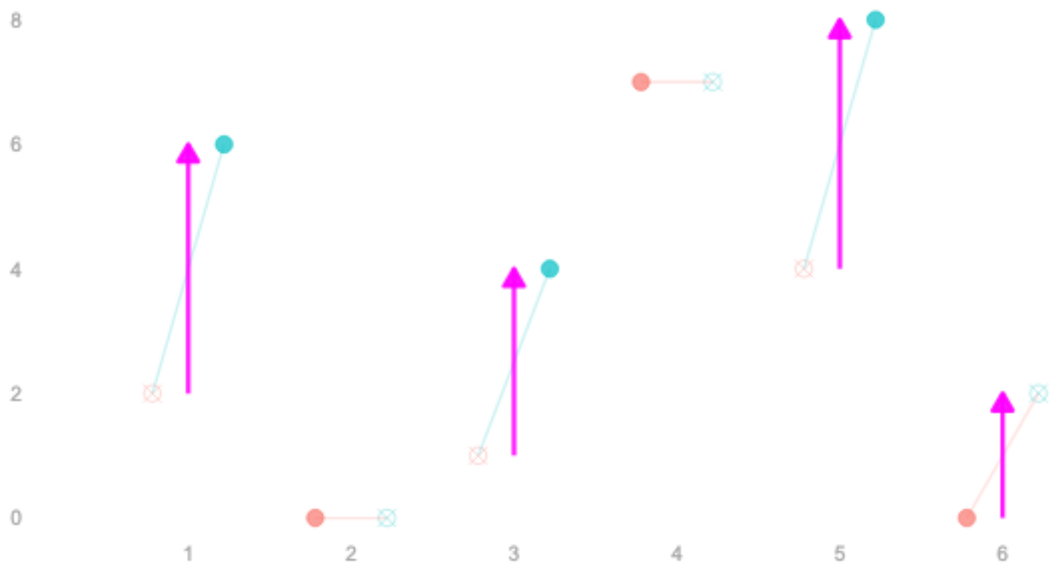


j	$y_j(1)$	$y_j(0)$	τ_j	w_j	$y_j(w_j)$
1	6	2		1	
2	0	0		0	
3	4	1		1	
4	7	7		0	
5	8	4		1	
6	2	0		0	

- Look at the treatments w_j in the table above. For each individual:
  - → Fill in the realized outcome $y_j(w_j)$ for each row in the table.
  - → Shade in the dot corresponding to the realized outcome in the plot.
  - → Draw arrows to indicate the individual treatment effects $\tau_j = y_j(1) - y_j(0)$ in the plot.
    - ⇾ Make them point up for positive effects, down for negative effects.
- Step forward to see the solution.

Initialise

The Fundamental Problem of Causal Inference



j	$y_j(1)$	$y_j(0)$	τ_j	w_j	$y_j(w_j)$
1	6	2	4	1	6
2	0	0	0	0	0
3	4	1	3	1	4
4	7	7	0	0	7
5	8	4	4	1	8
6	2	0	2	0	0

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  - → Fill in the realized outcome $y_j(w_j)$ for each row in the table.
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    - ⇾ Make them point up for positive effects, down for negative effects.
- Step forward to see the solution.

Initialise

What does the ATE tell us about Individual Effects?

What does an average treatment effect tell you, and what does it not?

- We've just defined $\bar{\tau}$. Before we ask whether we can estimate it, let's see how little it pins down.
- Randomization implies that the *probability distribution* of both potential outcomes is the same in both groups.
- Let's take this to an extreme.
  - → We'll pretend it implies the *observed distribution* (i.e. the set of values) is the same.
  - → All we get to fiddle with is *who* we give each value to.
- Let's make this a puzzle.
  - → The empty $y_j(1)$ cells for $j = 4, 5, 6$ must be filled with a 4, a 3, and a 2 (in some order)...
  - → And the empty $y_j(0)$ cells for $j = 1, 2, 3$ must be filled with a 3, a 2, and a 1 (in some order).

Find two solutions:

- with constant treatment effects
- with both negative and positive effects

j	w_j	$y_j(w_j)$	$y_j(1)$	$y_j(0)$	τ_j
1	1	4	4		
2	1	3	3		
3	1	2	2		
4	0	3		3	
5	0	2		2	
6	0	1		1	

j	w_j	$y_j(w_j)$	$y_j(1)$	$y_j(0)$	τ_j
1	1	4	4		
2	1	3	3		
3	1	2	2		
4	0	3		3	
5	0	2		2	
6	0	1		1	

Initialise

What does the ATE tell us about Individual Effects?

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j	w_j	$y_j(w_j)$	$y_j(1)$	$y_j(0)$	τ_j
1	1	4	4	3	1
2	1	3	3	2	1
3	1	2	2	1	1
4	0	3	4	3	1
5	0	2	3	2	1
6	0	1	2	1	1

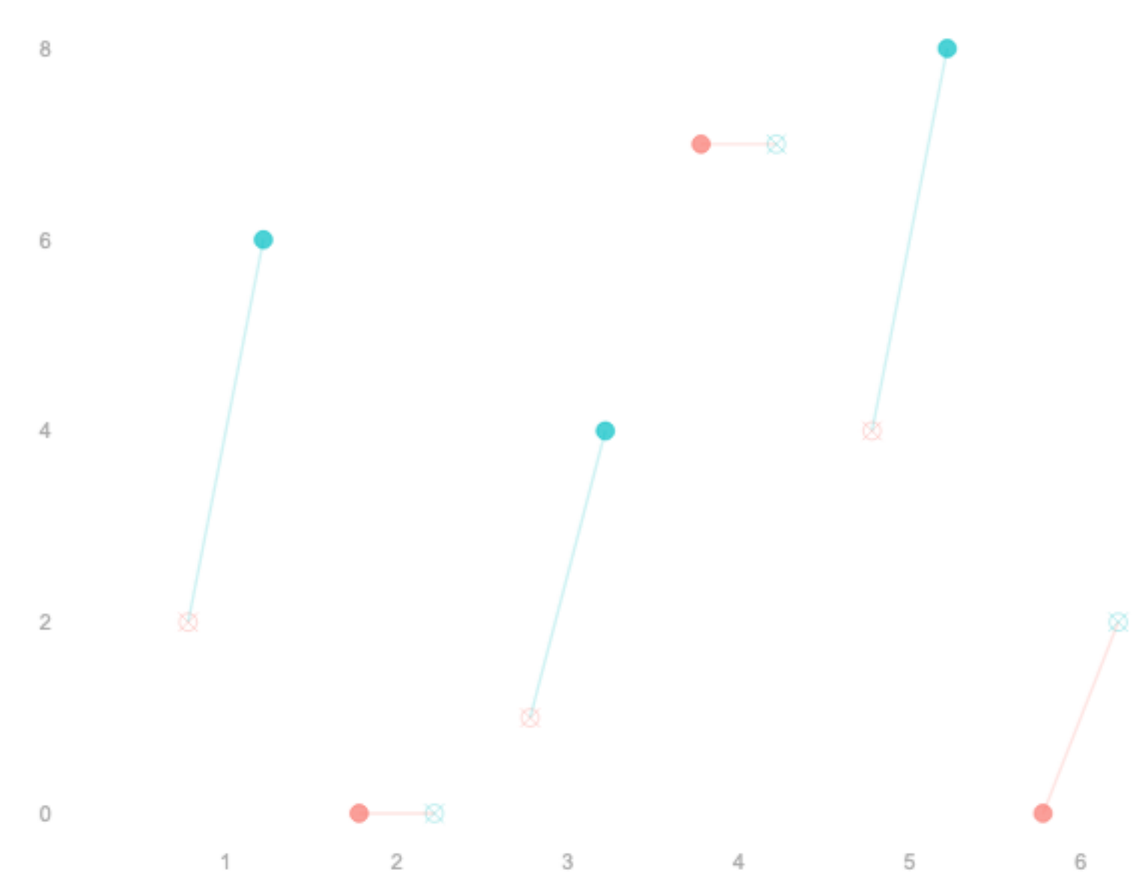
j	w_j	$y_j(w_j)$	$y_j(1)$	$y_j(0)$	τ_j
1	1	4	4	1	3
2	1	3	3	2	1
3	1	2	2	3	−1
4	0	3	2	3	−1
5	0	2	3	2	1
6	0	1	4	1	3

Initialise

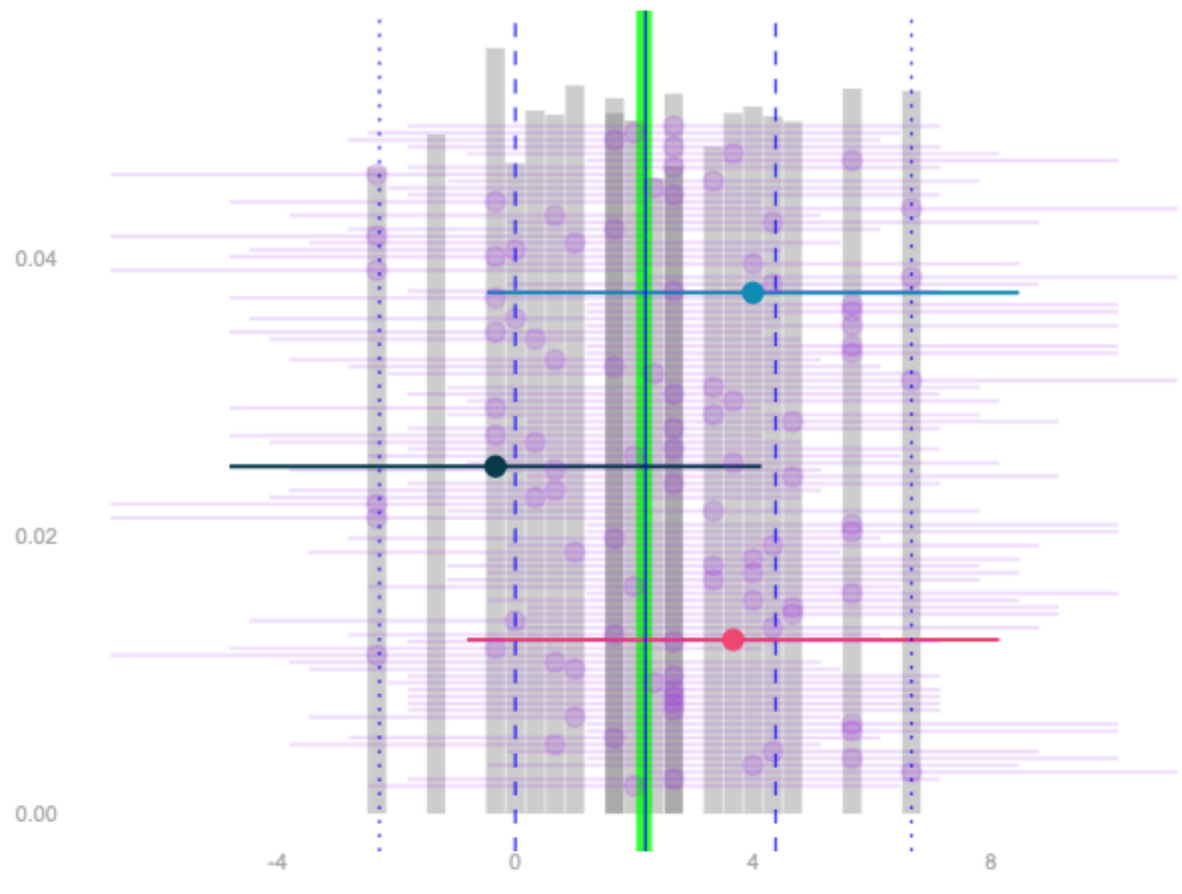
Randomization and Its Implications

Initialise

The Fundamental Solution is Randomization



The realized outcomes after randomizing treatment (once)



The sampling distribution when we randomize treatment (over and over)

- Step 1. *Randomize* treatment.
  - → Shuffle a deck of m cards labeled $1 \ldots m$.
  - → Treat individual j if the card labeled j is in the top m_1 cards of the deck, leaving $m_0 = m - m_1$ in control.
- Step 2. *Compare* the mean realized outcome for groups receiving different treatments.

$$\hat{\tau} = \frac{1}{m_1} \sum_{j:W_j=1} y_j(W_j) - \frac{1}{m_0} \sum_{j:W_j=0} y_j(W_j)$$

- Above, we see the sampling distribution[2] of this estimator $\hat{\tau}$ with all the usual annotations.

Initialise

Formalizing Shuffling

- We've said this before.
- The most common randomization technique is sampling the treated group without replacement.
  - → It's sometimes called **complete randomization**.
  - → In physical terms, you shuffle a deck and pull off the top however many you want.
  - → Those are the indices j of your treated people.
- Let's formalize this.
  - → Let $D_1 \dots D_m$ be the depths of cards $1 \dots m$ post-shuffle.
  - → A perfect shuffle makes every permutation equally likely.
  - → Each sequence has probability 1 over the number of possible sequences, $m(m-1)\dots 1$.
  - → Whether someone lands in the treated group is a function of D_j.

$$W_j = 1_{\leq m_1}(D_j)$$

  - → Here m_1 is the number of treated units you're going to have.
- What do we know about W_j?
  - → Today we want one thing only: its marginal probability.
- What is that?
  - → *Hint*. By symmetry, card j is equally likely to land at any depth $1 \dots m$.
  - → How many of those m depths put person j in the treated group?

Initialise

Formalizing Shuffling

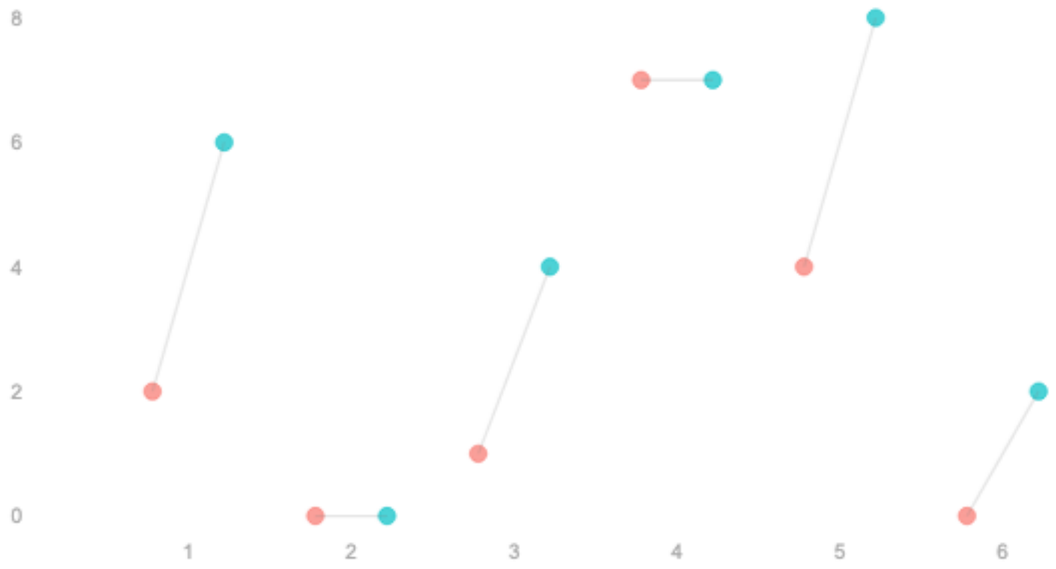
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- What is that?
  - → *Hint*. By symmetry, card j is equally likely to land at any depth $1 \dots m$.
  - → How many of those m depths put person j in the treated group?
- Since all m depths are equally likely and m_1 of them are in the top m_1,

Initialise

The Indicator Trick

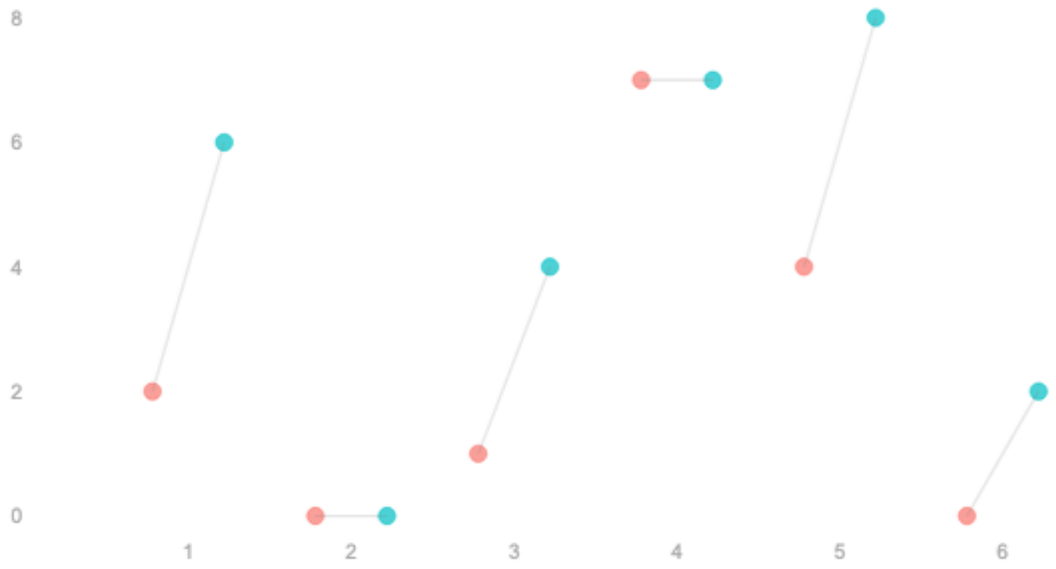


j	$y_j(1)$	$y_j(0)$	τ_j
1	6	2	4
2	0	0	0
3	4	1	3
4	7	7	0
5	8	4	4
6	2	0	2

- Suppose we randomize treatment W_j for person j in the population of 6 people above.
- What is $1_{=1}(W_j)y_j(W_j)$?

Initialise

The Indicator Trick



j	$y_j(1)$	$y_j(0)$	τ_j
1	6	2	4
2	0	0	0
3	4	1	3
4	7	7	0
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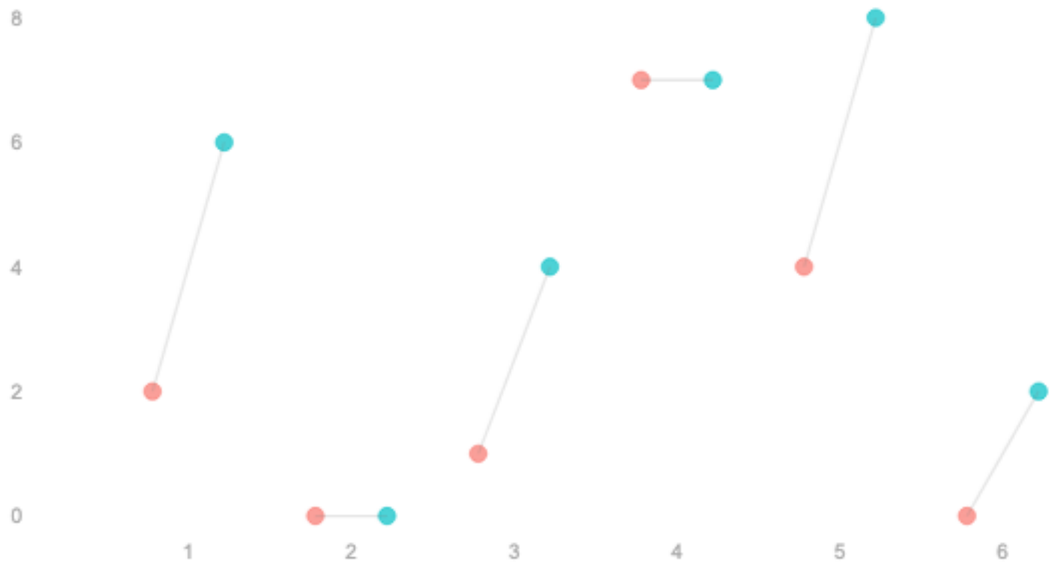
- $1_{=1}(W_j)y_j(W_j)$ is a random variable taking on these two values.

$$1_{=1}(W_j)y_j(W_j) = \begin{cases} 0 \times y_j(0) & \textbf{when } W_j = 0 \\ 1 \times y_j(1) & \textbf{when } W_j = 1 \end{cases}$$

- We can write it equivalently as $1_{=1}(W_j)y_j(1)$ because either ...
  - → $W_j = 0$, so $1_{=1}(W_j)y_j(W_j) = 1_{=1}(W_j)y_j(1) = 0$
  - → $W_j = 1$, so $1_{=1}(W_j)y_j(W_j) = 1_{=1}(W_j)y_j(1) = y_j(1)$

Initialise

The Indicator Trick



j	$y_j(1)$	$y_j(0)$	τ_j
1	6	2	4
2	0	0	0
3	4	1	3
4	7	7	0
5	8	4	4
6	2	0	2

- Suppose we randomize treatment W_j for person j in the population of 6 people above.
- What is $1_{=1}(W_j)y_j(W_j)$?

- More generally, for any function f,

$$1_{=w}(W)f(W) = \begin{cases} f(w) & \text{when } W = w \\ 0 & \text{when } W \neq w \end{cases} = 1_{=w}(W)f(w)$$

- We call this the **indicator trick**.
  - → It comes up a lot with subsample means.
- Shuffling tells us one thing about W_j.
  - → Its marginal probability is $\mathrm{P}(W_j = w) = m_w/m$.
  - → The trick moves the randomness into the indicator.

Initialise

How to Write Subsample Means

- We can write a subsample mean as sums over the whole population, top and bottom …
  - → in the numerator, the sum of outcomes in the group.
  - → in the denominator, the number of those terms.
  - → That's a *sum of ones* over the group.

$$\hat{\mu}(1) = \frac{\sum_{j:W_j=1} y_j(W_j)}{\sum_{j:W_j=1} 1} = \frac{\sum_{j=1}^{m} 1_{=1}(W_j) y_j(W_j)}{\sum_{j=1}^{m} 1_{=1}(W_j)}$$

- It's easy to make mistakes using *linearity of expectations* when we're summing over a subpopulation.
  - → Can we or can we not 'push' or 'pull' an expectation through a sum …
  - → … when the terms in that sum depend on the value of a random variable?
- To make this a lot more obvious, we can rewrite these as sums over the whole population.
  - → Instead of 'excluding' the terms where $W_j = 0$ …
  - → … we 'make them zero' by multiplying in the indicator $1_{=1}(W_j)$.
  - → Then, of course we can distribute expectations through the sum.

Initialise

Proving Unbiasedness

- Take the expectation of one group mean, then of the difference.

$$\begin{aligned}
\hat{\mu}(w) &\stackrel{?}{=} \frac{1}{m_w} \sum_{j:W_j=w} y_j(W_j) \\
&\stackrel{?}{=} \frac{1}{m_w} \sum_{j=1}^{m} y_j(W_j)\ 1_{=w}(W_j) \\
&\stackrel{?}{=} \frac{1}{m_w} \sum_{j=1}^{m} y_j(w) 1_{=w}(W_j).
\end{aligned}$$

$$\begin{aligned}
\mathrm{E}[\hat{\mu}(w)] &\stackrel{?}{=} \frac{1}{m_w} \sum_{j=1}^{m} y_j(w)\ \mathrm{E}[1_{=w}(W_j)] \\
&\stackrel{?}{=} \frac{1}{m_w} \sum_{j=1}^{m} y_j(w)\ \frac{m_w}{m} \\
&\stackrel{?}{=} \frac{1}{m} \sum_{j=1}^{m} y_j(w) \\
\mathrm{E}[\hat{\tau}] &\stackrel{?}{=} \mathrm{E}[\hat{\mu}(1)] - \mathrm{E}[\hat{\mu}(0)] \\
&\stackrel{?}{=} \frac{1}{m} \sum_{j=1}^{m} y_j(1) - \frac{1}{m} \sum_{j=1}^{m} y_j(0) \\
&\stackrel{?}{=} \frac{1}{m} \sum_{j=1}^{m} \{y_j(1) - y_j(0)\} = \bar{\tau}.
\end{aligned}$$

Initialise

Dropping Treatment Groups



ECRLOT **C002
THE JONES FAMILY
9999 WILLIAMS RD
FLINT MI 48507

Dear Registered Voter:

DO YOUR CIVIC DUTY AND VOTE!

Why do so many people fail to vote? We've been talking about this problem for years, but it only seems to get worse.

The whole point of democracy is that citizens are active participants in government; that we have a voice in government. Your voice starts with your vote. On August 8, remember your rights and responsibilities as a citizen. Remember to vote.

DO YOUR CIVIC DUTY — VOTE!



30424-1

For more information: (517) 351-1975
email: etow@grebner.com
Practical Political Consulting
P. O. Box 6249
East Lansing, MI 48826

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ECRLOT **C001
THE SMITH FAMILY
9999 PARK LANE
FLINT MI 48507

Dear Registered Voter:

YOU ARE BEING STUDIED!

Why do so many people fail to vote? We've been talking about this problem for years, but it only seems to get worse.

This year, we're trying to figure out why people do or do not vote. We'll be studying voter turnout in the August 8 primary election.

Our analysis will be based on public records, so you will not be contacted again or disturbed in any way. Anything we learn about your voting or not voting will remain confidential and will not be disclosed to anyone else.

DO YOUR CIVIC DUTY — VOTE!



ECRLOT **C060
THE WAYNE FAMILY
9999 OAK ST
FLINT MI 48507

Dear Registered Voter:

WHO VOTES IS PUBLIC INFORMATION!

Why do so many people fail to vote? We've been talking about the problem for years, but it only seems to get worse.

This year, we're taking a different approach. We are reminding people that who votes is a matter of public record.

The chart shows your name from the list of registered voters, showing past votes, as well as an empty box which we will fill in to show whether you vote in the August 8 primary election. We intend to mail you an updated chart when we have that information.

We will leave the box blank if you do not vote.

DO YOUR CIVIC DUTY—VOTE!

OAK ST	Aug 04	Nov 04	Aug 06
9999 ROBERT WAYNE		Voted	
9999 LAURA WAYNE	Voted	Voted	



Dear Registered Voter:

WHAT IF YOUR NEIGHBORS KNEW WHETHER YOU VOTED?

Why do so many people fail to vote? We've been talking about the problem for years, but it only seems to get worse. This year, we're taking a new approach. We're sending this mailing to you and your neighbors to publicize who does and does not vote.

The chart shows the names of some of your neighbors, showing which have voted in the past. After the August 8 election, we intend to mail an updated chart. You and your neighbors will all know who voted and who did not.

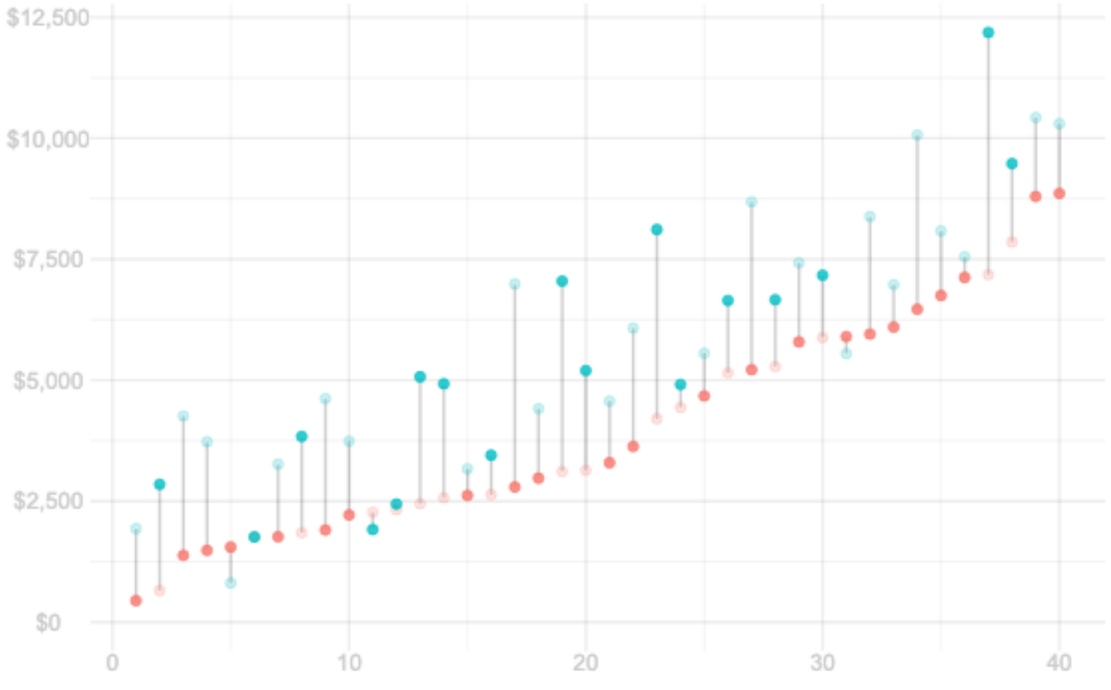
DO YOUR CIVIC DUTY—VOTE!

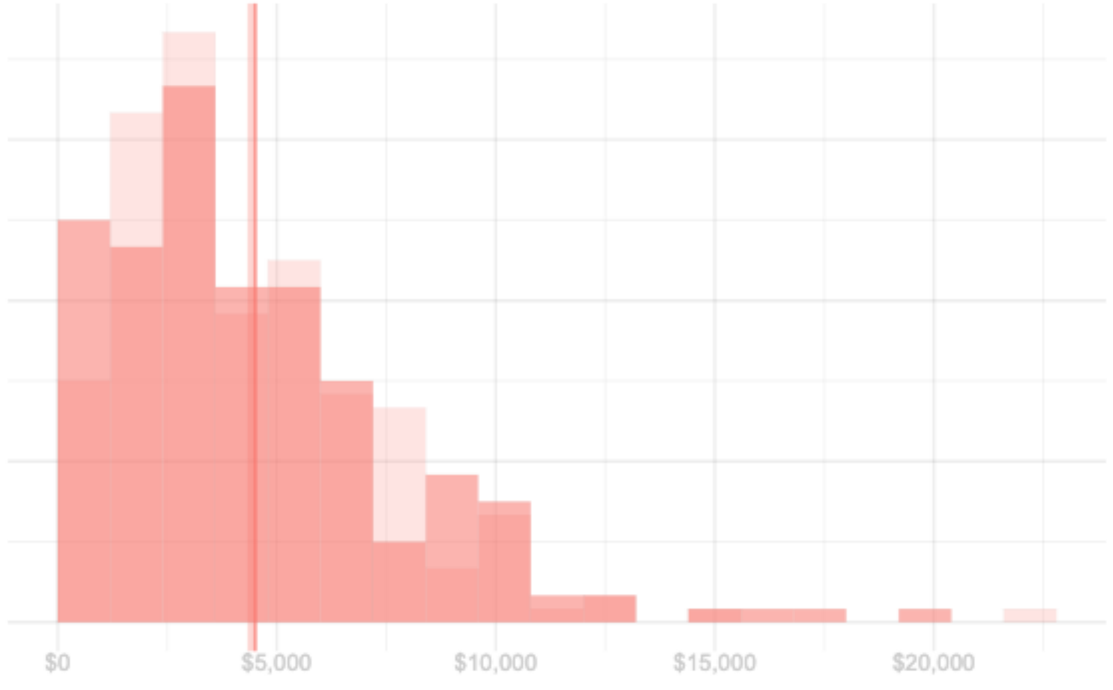
MAPLE DR	Aug 04	Nov 04	Aug 06
9985 JOSEPH JAMES SMITH	Voted	Voted	
9985 JENNIFER KAY SMITH		Voted	
9997 RICHARD B JACKSON		Voted	
9999 KATHY MARIE JACKSON		Voted	
9999 BRIAN JOSEPH JACKSON		Voted	
9991 JENNIFER KAY THOMPSON		Voted	
9991 BOB R THOMPSON		Voted	
9993 BILL S SMITH			
9989 WILLIAM LUKE CASPER		Voted	
9989 JENNIFER SUE CASPER		Voted	
9987 MARIA S JOHNSON	Voted	Voted	
9987 TOM JACK JOHNSON	Voted	Voted	
9987 RICHARD TOM JOHNSON		Voted	
9985 ROSEMARY S SUE		Voted	
9985 KATHRYN L SUE		Voted	
9985 HOWARD BEN SUE		Voted	
9983 NATHAN CHAD BERG		Voted	

- In our first lecture, the Michigan experiment had five letters, $w = 0 \ldots 4$.
- We tossed out the three groups we weren't analyzing.
- That left Neighbors ($w = 4$) against no letter ($w = 0$).
  - → The four letters that went out were Civic Duty, Hawthorne, Self and Neighbors. The fifth arm got nothing.
- This is the same cancellation we just did, with five treatments instead of two.
- Under shuffling with fixed group sizes $m_0 \ldots m_4$, tossing out the groups you aren't analyzing gives the exact same target as if you'd only included those two.
  - → Each group mean $\hat{\mu}(w) = \frac{1}{m_w} \sum_{j=1}^{m} y_j(w) 1_{=w}(W_j)$ has $\mathbf{E}[1_{=w}(W_j)] = m_w/m$.
  - → The m_w in the denominator cancels with m_w/m.
  - → So $\mathbf{E}[\hat{\mu}(4) - \hat{\mu}(0)] = \frac{1}{m} \sum_{j=1}^{m} \{y_j(4) - y_j(0)\}$.

Initialise

What We See and What We Don’t

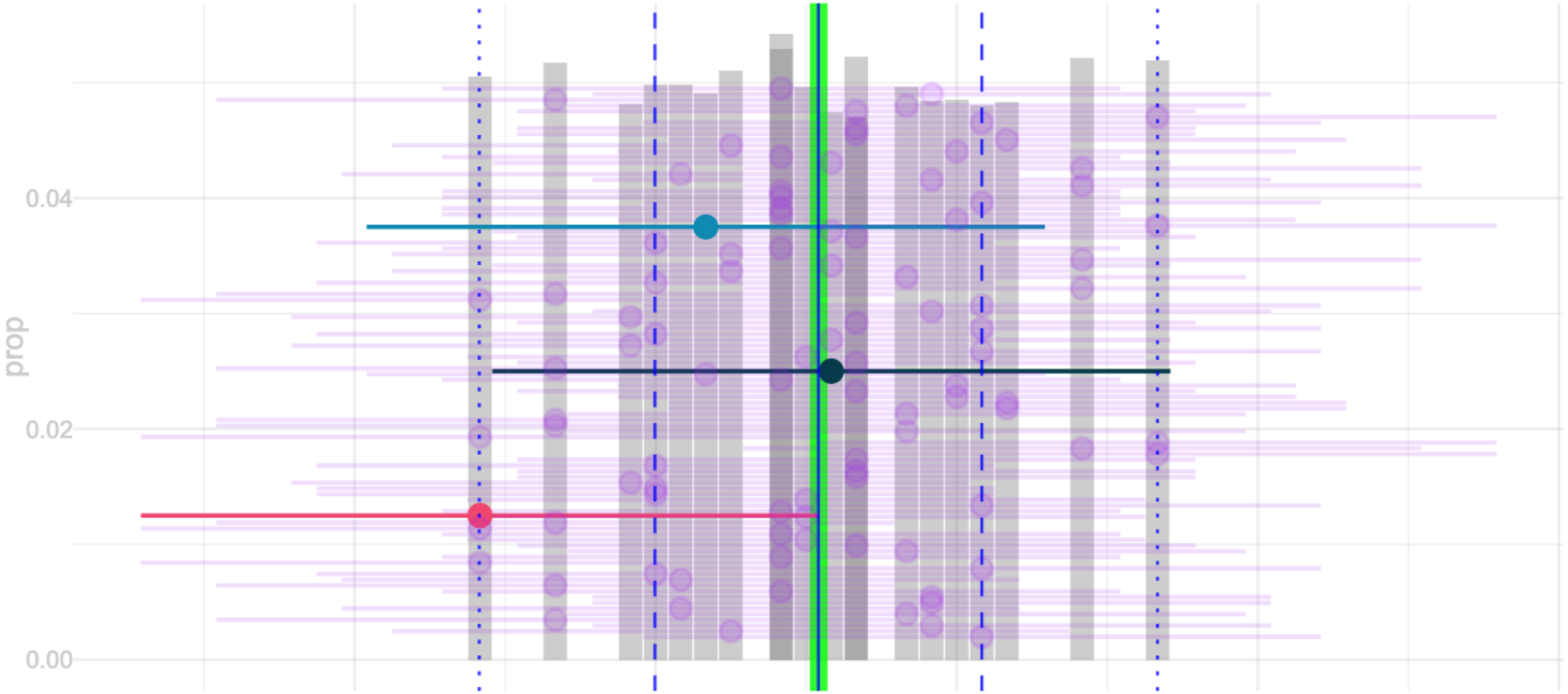




- Faded is the control outcome of someone we treated. We never see it.
- Randomization is what makes it look like the control outcomes we do see.

Initialise

That’s What We Got Today



- We know $\hat{\tau}$ is unbiased.
- The centre is in the right place.
- We haven’t worked out the spread.
- There are no numbers on that axis yet.
- That’s next time.

Initialise

When Does Someone Have Two Potential Outcomes?

Initialise

Texts I Got Around Election Day

Text Message • SMS
Yesterday 9:26 AM

It's Election Day, David! Vote by 7:00 pm for Jon Ossoff. Confirm polling place: demswin.us/go

Voter Hotline/Questions: 866-687-8683

ForwardBlue
End2End

Text Message • SMS
Yesterday 12:35 PM



Georgia, it's Keisha. Today, we choose the future our families deserve: lower costs, better health care, and strong schools. This race is close, and your vote can decide it. Together, we can make history. Polls are open until 7 PM. Find your assigned polling place, and bring your photo ID: mvp.sos.ga.gov

I also wanted to share Keisha Lance Bottoms' plan for Georgia's future. Take a look: https://online.flippingbook.com/view/699781803/

Text STOP to quit



Hi David, it's Mo Ivory for Fulton County Commission Chair.

Election Day is TOMORROW, Tuesday, November 3! This is your last chance to cast your ballot in this election, and every vote matters.

Here's what I need you to do:

1️⃣ Make a plan. Decide what time you're voting tomorrow. Polls are open 7 AM–7 PM.
2️⃣ Bring a friend. Call your family and remind them to vote, too. Don't assume everyone has a plan.
3️⃣ Verify your polling location. Our records show it's:

📍 PIEDMONT PARK MAGNOLIA HALL, 1320 MONROE DR NE, ATLANTA



Watch Video msg from Jodi Merriday. ⏰ EARLY VOTING ENDS AT 7. 🗳️ Elect Jodi Merriday Fulton County Commissioner of District 3. Decreasing 💵 financial burdens is a top goal! Citizens deserve a return on investment from government.

My platform is simple:

✅ I support secure, fair elections and the protection of every ballot.
✅ I support investment in public safety and justice partnerships to reduce crime.
✅ I believe hospitals, mental health, and senior services should be accessible and funded.
✅ I support public-private partnerships that reduce the fiscal burdens of everyday people.

Let's move Fulton County forward together! Election Day is Tuesday, November 3, with polls open from 7 a.m. to 7 p.m.

Voting information: mvp.sos.ga.gov (http://mvp.sos.ga.gov/)

Fulton County Elections: 404-612-7020

Learn more: www.jodimerriday.com (http://www.jodimerriday.com/)

Reply STOP to opt out.

Text Message • SMS
Monday 12:31 PM

Rick Jackson backed a $1 TRILLION cut to Medicaid, while giving record-breaking tax breaks to billionaires and big corporations instead. Medicaid covers 50% of seniors in nursing homes and 60% of kids when

Text Message • SMS
Yesterday 12:08 PM

Updated Voting Information.
Join GA law enforcement and Gov. Kemp in voting for Brian Strickland for Attorney General.
Vote Here: gop-txt.org/sv13xn

Text Message • SMS
Yesterday 10:27 AM

Polls are open until 7pm
Join GA law enforcement and Gov. Kemp in voting for Brian Strickland for Attorney General.
Vote Here: gop-txt.org/3fawtt

Text Message • SMS
Monday 3:50 PM

Endless wars = endless price increases in Georgia. Gas prices in Georgia have already increased by 33% a gallon since the start of the Iran war. Can you afford to keep paying higher prices at the pump this

The Ones That Read Like the Letters

Text Message • SMS
Yesterday 11:21 AM

David, VOTE BY TONIGHT. Election Day is TODAY, 11/3. Who you vote for is private, but whether you vote is public information.
-OurBallot
Stop to end

Text Message • SMS
Sunday 11:32 AM

Hi David, computer models say the likelihood of you voting Tuesday is: LIKELY. Public records show you voted in both 2022 and 2024. Find your voting info: https://sos.ga.gov/voter-resources
-NextVote
Stop to end

Text Message • SMS
Yesterday 9:56 AM

Fulton County Voters: LAST CHANCE TO VOTE! Today, Tues 11/3 is Election Day. Thousands of voters in Cherokee County already voted. Make sure Fulton County shows up too! Public records show you voted in both 2022 and 2024. Vote if you haven't!
-NextVote
Stop to end

Text Message • SMS
Monday 8:20 AM

David, public records show you voted in both 2022 and 2024. Make a plan to vote TOMORROW, 11/3.
-NextVote
Stop to end

Text Message • SMS
Yesterday 1:35 PM

LAST CHANCE TO VOTE: David, TODAY, Tuesday 11/3, is Election Day! Make sure to vote if you haven't already!
-NextVote
Stop to end

Text Message • SMS
Yesterday 9:14 AM

David, TODAY 11/3 IS ELECTION DAY! Vote now: https://sos.ga.gov/voter-resources
-OurBallot
Stop to end

Text Message • SMS
Monday 3:04 PM

One of Each, Side by Side



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Election Day is TOMORROW, Tuesday, November 3! This is your last chance to cast your ballot in this election, and every vote matters.

Here's what I need you to do:

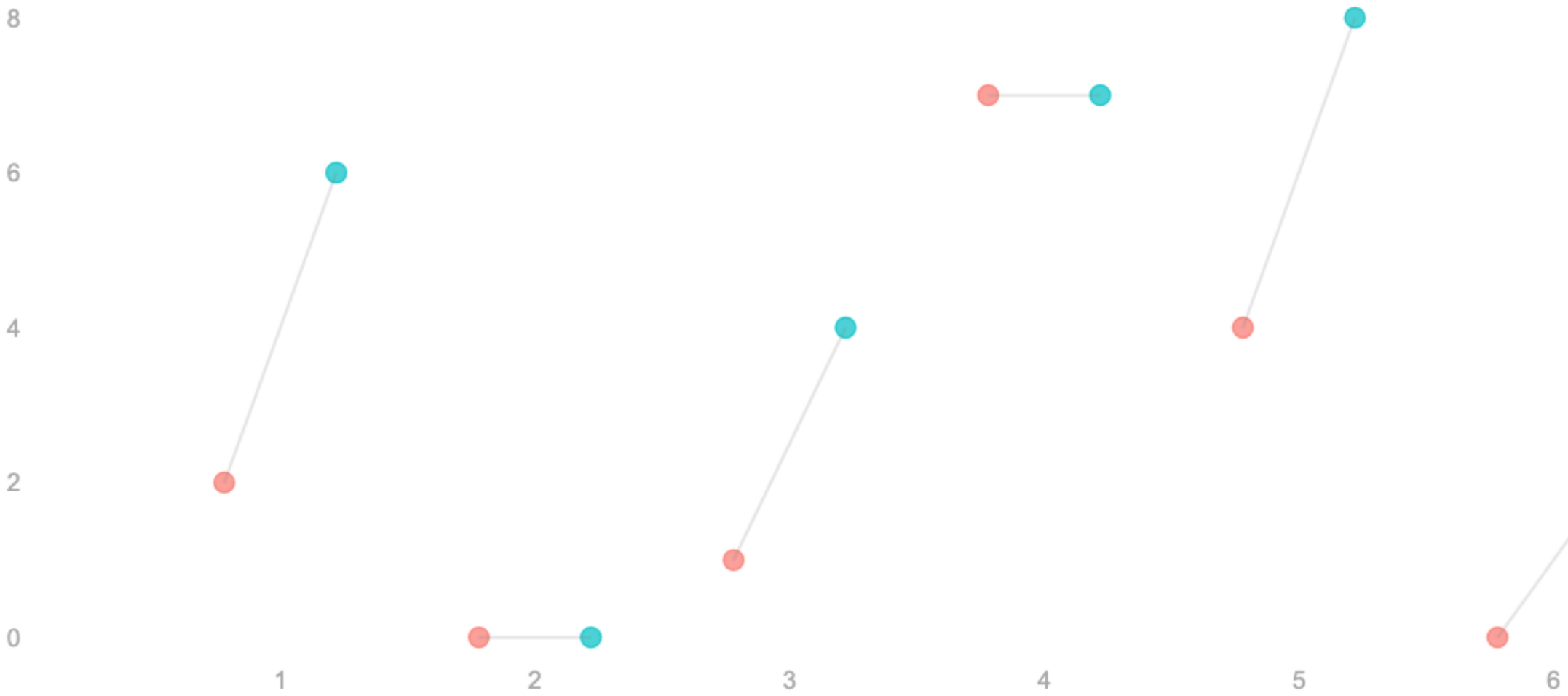
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2️⃣ Bring a friend. Call your family and remind them to vote, too. Don't assume

Text Message • SMS
Sunday 11:32 AM

Hi David, computer models say the likelihood of you voting Tuesday is: LIKELY. Public records show you voted in both 2022 and 2024. Find your voting info: https://sos.ga.gov/voter-resources
-NextVote
Stop to end

Initialise

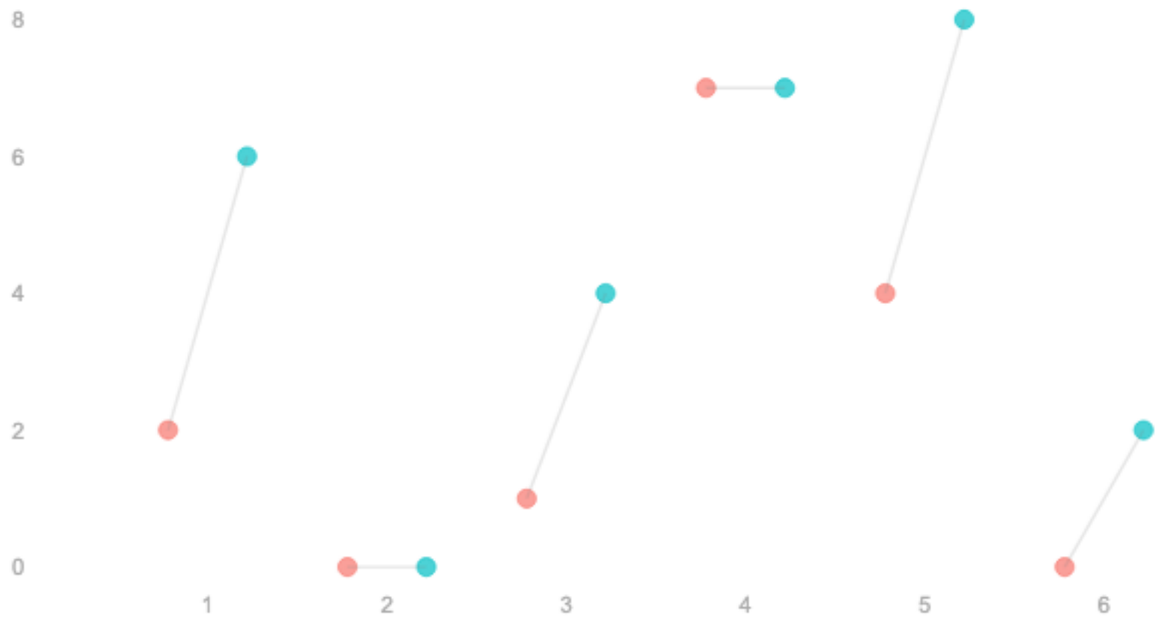
What Does Each End of the Pair Mean?



- Everything so far started by writing down a pair, $y_j(0)$ and $y_j(1)$, for each person.
  - → That's what each connected pair of dots above is.
- Writing the pair down is easy.
- Saying what each end *means* is the hard part.
- *Question*. For each comparison coming up, what would the other end of that person's pair be?
  - → If we can't say, we don't have a τ_j to average.

Initialise

Georgia: Black vs. non-Black Voters





- What is the ‘non-Black version’ of a Black person?
- Which specific counterfactual would we even mean?
- ‘Well-defined’ is math for ‘you actually know what you mean when you say it’.
- Eddie Murphy, *White Like Me*, SNL 1984: same man, makeup and a different voice, everything else held fixed.
- Bertrand and Mullainathan, *Are Emily and Greg More Employable Than Lakisha and Jamal? A Field Experiment on Labor Market Discrimination*, swap a name on a résumé.
- Kohler-Hausmann, *Eddie Murphy and the Dangers of Counterfactual Causal Thinking About Detecting Racial Discrimination*, asks what that measured.

Initialise

Gerber, Green and Larimer: the Letters



ECRLOT **C002
THE JONES FAMILY
9999 WILLIAMS RD
FLINT MI 48507

Dear Registered Voter:

DO YOUR CIVIC DUTY AND VOTE!

Why do so many people fail to vote? We've been talking about this problem for years, but it only seems to get worse.

The whole point of democracy is that citizens are active participants in government, that we have a voice in government. Your voice starts with your vote. On August 8, remember your rights and responsibilities as a citizen. Remember to vote.

DO YOUR CIVIC DUTY — VOTE!



30424-1

For more information: (517) 351-1975
email: etov@gerber.com
Practical Political Consulting
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ECRLOT **C001
THE SMITH FAMILY
9999 PARK LANE
FLINT MI 48507

Dear Registered Voter:

YOU ARE BEING STUDIED!

Why do so many people fail to vote? We've been talking about this problem for years, but it only seems to get worse.

This year, we're trying to figure out why people do or do not vote. We'll be studying voter turnout in the August 8 primary election.

Our analysis will be based on public records, so you will not be contacted again or disturbed in any way. Anything we learn about your voting or not voting will remain confidential and will not be disclosed to anyone else.

DO YOUR CIVIC DUTY — VOTE!



ECRLOT **C050
THE WAYNE FAMILY
9999 OAK ST
FLINT MI 48507

Dear Registered Voter:

WHO VOTES IS PUBLIC INFORMATION!

Why do so many people fail to vote? We've been talking about the problem for years, but it only seems to get worse.

This year, we're taking a different approach. We are reminding people that who votes is a matter of public record.

The chart shows your name from the list of registered voters, showing past votes, as well as an empty box which we will fill in to show whether you vote in the August 8 primary election. We intend to mail you an updated chart when we have that information.

We will leave the box blank if you do not vote.

DO YOUR CIVIC DUTY—VOTE!

OAK ST	Aug 04	Nov 04	Aug 06
9999 ROBERT WAYNE		Voted	
9999 LAURA WAYNE	Voted	Voted	



Dear Registered Voter:

WHAT IF YOUR NEIGHBORS KNEW WHETHER YOU VOTED?

Why do so many people fail to vote? We've been talking about the problem for years, but it only seems to get worse. This year, we're taking a new approach. We're sending this mailing to you and your neighbors to publicize who does and does not vote.

The chart shows the names of some of your neighbors, showing which have voted in the past. After the August 8 election, we intend to mail an updated chart. You and your neighbors will all know who voted and who did not.

DO YOUR CIVIC DUTY—VOTE!

MAPLE DR	Aug 04	Nov 04	Aug 06
9985 JOSEPH JAMES SMITH	Voted	Voted	
9985 JENNIFER KAY SMITH		Voted	
9997 RICHARD B JACKSON		Voted	
9999 KATHY MARIE JACKSON		Voted	
9999 BRIAN JOSEPH JACKSON		Voted	
9991 JENNIFER KAY THOMPSON		Voted	
9991 BOB R THOMPSON		Voted	
9993 BILL S SMITH			
9989 WILLIAM LUKE CASPER		Voted	
9989 JENNIFER SUE CASPER		Voted	
9987 MARIA S JOHNSON	Voted	Voted	
9987 TOM JACK JOHNSON	Voted	Voted	
9987 RICHARD TOM JOHNSON		Voted	
9985 ROSEMARY S SUE		Voted	
9985 KATHRYN L SUE		Voted	
9985 HOWARD BEN SUE		Voted	
9983 NATHAN CHAD BERG		Voted	

- The letter went out, and it could just as easily not have.
- So the other end of the pair is well-defined.
- We know exactly what we would have done differently.
  - → That's not the same as knowing the turnout it would have produced.
  - → We never see that.

Initialise

The Texts: Is There Another Version of Me?



Hi David, it's Mo Ivory for Fulton County Commission Chair.

Election Day is TOMORROW, Tuesday, November 3! This is your last chance to cast your ballot in this election, and every vote matters.

Here's what I need you to do:

1️⃣ Make a plan. Decide what time you're voting tomorrow. Polls are open 7 AM–7 PM.
2️⃣ Bring a friend. Call your family and remind them to vote, too. Don't assume

Text Message • SMS
Sunday 11:32 AM

Hi David, computer models say the likelihood of you voting Tuesday is: LIKELY. Public records show you voted in both 2022 and 2024. Find your voting info: https://sos.ga.gov/voter-resources
-NextVote
Stop to end

Initialise

What Each One Would Tell Us



Dear Registered Voter:

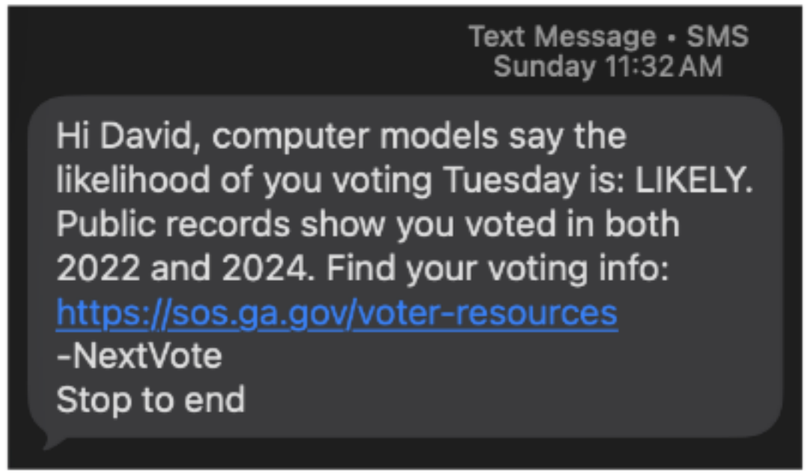
WHAT IF YOUR NEIGHBORS KNEW WHETHER YOU VOTED?

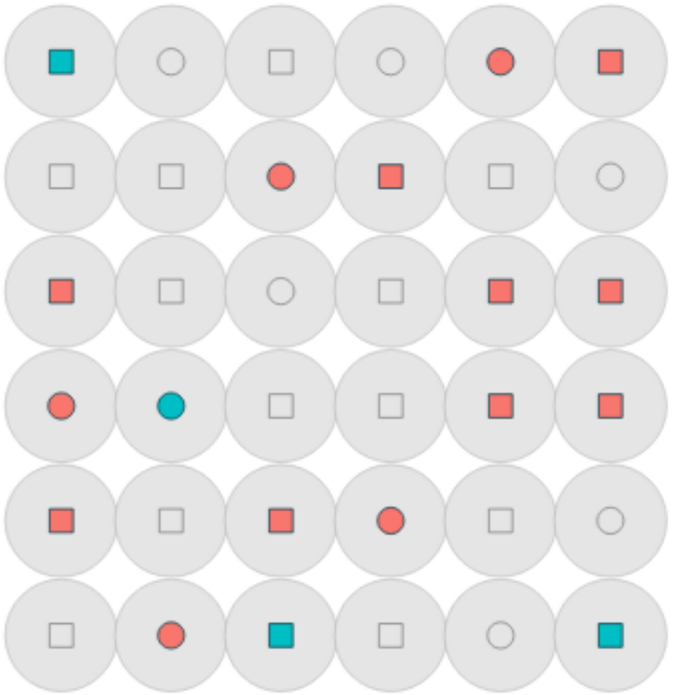
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The chart shows the names of some of your neighbors, showing which have voted in the past. After the August 8 election, we intend to mail an updated chart. You and your neighbors will all know who voted and who did not.

DO YOUR CIVIC DUTY — VOTE!

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9995 JENNIFER KAY SMITH		Voted	_____
9997 RICHARD B JACKSON		Voted	_____
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9991 JENNIFER KAY THOMPSON		Voted	_____
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9993 BILL S SMITH			_____
9989 WILLIAM LUKE CASPER		Voted	_____
9989 JENNIFER SUE CASPER		Voted	_____
9987 MARIA S JOHNSON	Voted	Voted	_____
9987 TOM JACK JOHNSON	Voted	Voted	_____
9987 RICHARD TOM JOHNSON		Voted	_____
9985 ROSEMARY S SUE		Voted	_____
9985 KATHRYN L SUE		Voted	_____
9985 HOWARD BEN SUE		Voted	_____
9983 NATHAN CHAD BERG		Voted	





- Compare three estimates $\hat{\Delta} = \hat{\mu}(1) - \hat{\mu}(0)$.
  1. Neighbors letter vs. no letter in Gerber, Green, and Larimer

Initialise

Footnotes

1.

This is inconsistent with everyday use of the term *treatment* because you can, e.g., take both ibuprofen and acetaminophen. In potential outcomes language, we'd say that ibuprofen alone, acetaminophen alone, and ibuprofen plus acetaminophen are three different treatments.

2.

Some people prefer to say 'randomization distribution' in this context because we're not sampling individuals from our population—we get a dot for each one.

Initialise